

# NovaCart

## Shopping Cart using a Stateful Session Bean

Advanced Web Programming · CA3 · Batch 2 / Group 1

| Prepared by         | Technology                                   |
|---------------------|----------------------------------------------|
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| Academic submission | Payara Micro · Servlet · HTML/CSS/JavaScript |

A customer-specific shopping cart with a polished web interface, container-managed conversation state and tested add, view, remove and clear operations.

Submitted as an academic demonstration. Product prices are sample values; no payment or order is processed.

# 01 Title

NovaCart: Online Shopping Cart using an Enterprise JavaBean Stateful Session Bean.

# 02 Aim

To develop a Java application using a Stateful Session Bean that maintains an individual customer's shopping cart across multiple operations and displays its current contents.

# 03 Problem Statement

Accept a customer name and allow the customer to add Laptop, Mouse and Keyboard products. Provide add, view, remove and clear operations. Use @Stateful and demonstrate that products are remembered throughout the customer interaction.

Version 2 expands the original lab into a 12-product catalog with six categories, search, sorting, product details, a live header count and a dedicated cart page. The required Laptop, Mouse and Keyboard products remain available.

# 04 Objectives

Create a real container-managed Stateful EJB; preserve customer-specific state; integrate it with a servlet client; display cart contents and totals; validate inputs; demonstrate isolated customer conversations and bean lifecycle management.

| PDF requirement                   | Implemented evidence                          |
|-----------------------------------|-----------------------------------------------|
| Accept Customer Name              | Customer form → setCustomerName(name)         |
| Laptop, Mouse, Keyboard           | Server-owned Catalog.PRODUCTS                 |
| Retain products across operations | ShoppingCartBean.quantities map               |
| Add / View / Remove / Clear       | add(), viewCart(), remove(), clear()          |
| Use @Stateful                     | Annotation on ShoppingCartBean                |
| Display current cart              | Dedicated cart.html page from server snapshot |
| Demonstrate maintained state      | Refresh retention + separate browser sessions |

## 05 Software Requirements

| Requirement             | Selected setup                                         |
|-------------------------|--------------------------------------------------------|
| Java runtime / build    | JDK 17, Apache Maven 3.9+                              |
| EJB application server  | Payara Micro 6.2025.1, Jakarta EE 10                   |
| Frontend / browser      | HTML, CSS, JavaScript; current Chromium/Edge/Firefox   |
| Testing                 | JUnit Jupiter 5.11.4; Python 3 for optional HTTP tests |
| Hardware recommendation | Laptop/desktop, at least 4 GB RAM; 8 GB recommended    |
| Internet                | First download of build dependencies/runtime only      |

Plain Apache Tomcat is insufficient because it does not provide the required EJB container. The included launch scripts use a pinned Payara runtime with checksum verification and clustering disabled for a simple lab deployment.

## 06 Introduction to Stateful Session Bean

A Stateful Session Bean is an enterprise component whose instance retains conversational state for a particular client across business-method calls. The Jakarta EE container manages its creation, invocation, timeout, optional passivation and removal. `@Stateful` declares this behavior; it is not merely a Java object held in a global variable.

| Aspect       | Stateless Session Bean           | Stateful Session Bean                   |
|--------------|----------------------------------|-----------------------------------------|
| Annotation   | <code>@Stateless</code>          | <code>@Stateful</code>                  |
| Client state | No guarantee across calls        | Retained for a conversation             |
| Instances    | Usually pooled and reused        | Associated with a client conversation   |
| Typical use  | Independent calculation          | Shopping cart or multi-step workflow    |
| Removal      | Container-managed pool lifecycle | Explicit <code>@Remove</code> or expiry |

## 07 Why Stateful Bean is used in the application

Shopping is a sequence of related operations: a customer adds a laptop, views the cart, adds a mouse and removes a keyboard. These calls must use the previous selection. A Stateful Bean naturally represents this conversation and isolates it from another customer's cart.

State fields are `customerName` (String), `quantities` (LinkedHashMap of product IDs to integer quantities), `conversationId` (UUID string) and `revision` (long). Product prices belong to an immutable server catalog and are not customer state. `HttpSession` stores the EJB proxy plus a CSRF token; actual customer/cart data lives in the bean.

## 08 Architecture / Application Flow

index.html presents the catalog; cart.html presents the selected products, quantity controls and summary. Both pages share the same JavaScript client and invoke the same /api/cart endpoint with the same HTTP session cookie. The header count represents total selected units, so two units of one product count as two items.

| Layer                      | Responsibility                                            |
|----------------------------|-----------------------------------------------------------|
| Browser UI                 | Collect name and operations; display server snapshots     |
| CartServlet                | Validate request/token, serialize same-session operations |
| HttpSession                | Retain one EJB proxy and CSRF token per browser session   |
| ShoppingCartBean           | Maintain customer and selected quantities                 |
| Catalog / snapshot objects | Own prices; return immutable items and BigDecimal totals  |

On the first request, JNDI lookup of java:module/ShoppingCartBean obtains a new Stateful Bean. The servlet retains its proxy in the HTTP session. Later requests invoke that same proxy. A separate browser session obtains a different bean. Never inject one shared cart into a servlet field.

Flow: Browser sends session cookie → servlet retrieves EJB proxy → bean reads/updates state → immutable snapshot is serialized as JSON → browser redraws the cart. Ending the HTTP session triggers SessionCleanup, which invokes @Remove close(). HTTP inactivity expires after 30 minutes; the bean timeout is 35 minutes.

## 09 Stateful Bean Class Structure

```
@Stateful
@StatefulTimeout(value = 35, unit = TimeUnit.MINUTES)
public class ShoppingCartBean implements Serializable {
    private String customerName;
    private final String conversationId;
    private final Map<String, Integer> quantities;
    private long revision;
    // setCustomerName, add, updateQuantity, remove, clear, viewCart
    @Remove public void close() { ... }
}
```

Product, CartLine and CartSnapshot are serializable immutable records. Complete executable classes are included in the Source Code appendix and the ZIP project.

## 10 Algorithm / Program Steps

1. Deploy the WAR on Payara and open /novacart/.
2. Create an HTTP session and look up one Stateful EJB proxy if none exists.
3. Accept and validate a customer name; store it in the bean.
4. For add(), validate product ID, enforce the 10-unit limit and increment its quantity.
5. For viewCart(), read the same map and return immutable line items and totals.
6. For remove(), delete the entire selected product line.

7. For clear(), empty quantities but preserve customer identity and conversation ID.
8. Click the header cart link, adjust quantities on cart.html, refresh and verify the same products and conversation.
9. End session to invoke @Remove; the next interaction starts a new conversation.

## 11 Source Code

Complete authored application source appears in Appendix A: all Java classes, Maven configuration, deployment descriptor, HTML, CSS, JavaScript and launch scripts. The ZIP also contains JUnit tests, an HTTP integration test, screenshots and the built WAR. Long source lines are visually wrapped in this report; use original files from the ZIP for compilation.

## 12 Input

| Input                   | Value                                                  |
|-------------------------|--------------------------------------------------------|
| Customer Name           | Numan                                                  |
| Products added in order | Laptop, Mouse, Keyboard                                |
| Sample prices           | Laptop ₹64,990.00; Mouse ₹1,299.00; Keyboard ₹3,499.00 |
| Additional operation    | Add Mouse once more; remove Keyboard                   |

## 13 Output

| Operation              | Expected and verified cart state                   |
|------------------------|----------------------------------------------------|
| Add three products     | 3 units; total ₹69,788.00                          |
| Refresh browser        | Same customer, products and conversation ID        |
| Add Mouse again        | 4 units; Mouse quantity 2; total ₹71,087.00        |
| Remove Keyboard        | 3 units; total ₹67,588.00                          |
| Clear cart             | 0 units; ₹0.00; Numan and conversation ID retained |
| Second browser session | Independent empty cart, different conversation ID  |

Actual browser screenshots are provided in Appendix B and docs/. Conversation UUID values are runtime-generated and differ on every new session.

## 14 Explanation of Important Code

@Stateful tells the EJB container that each obtained instance belongs to a conversational client. quantities is a private instance field, not static, so one customer cannot share another customer's product selections.

`add(productId)` first requires a valid customer, validates the catalog ID, reads the previous quantity and stores quantity + 1. Only successful mutations increment revision. `viewCart()` reads the already stored map, creates `CartLine` values and returns an immutable `CartSnapshot`. `BigDecimal` computes totals without binary floating-point rounding.

`@ApplicationException(rollback = false)` marks `CartValidationException` as an expected application failure. Otherwise an unchecked validation exception may be treated as an EJB system exception and discard the Stateful Bean. This annotation keeps the conversation usable after invalid input.

`@Remove` on `close()` asks the container to destroy the bean after the method completes. `clear()` intentionally does not use `@Remove`, because clearing products should not end the customer interaction. Serializable state supports passivation; our test verifies serialization, not forced container passivation.

The servlet validates a CSRF header for every POST, owns product pricing on the server, disables JSON caching, and does not expose a mutable reference to the cart map. Frontend values from the server/customer use `textContent`, preventing HTML injection in names. Same-session operations run within `synchronized(session)`.

## 15 Test Cases

| Case | Input / operation               | Expected result                    |
|------|---------------------------------|------------------------------------|
| 1    | Add Laptop + Mouse + Keyboard   | 3 units, ₹69,788.00                |
| 2    | Refresh same browser            | Customer/items/ID retained         |
| 3    | Add Mouse twice                 | One Mouse line, quantity 2         |
| 4    | Remove Keyboard                 | Line removed; total recalculated   |
| 5    | Clear populated cart            | Items empty; name/ID unchanged     |
| 6    | Open isolated browser session   | Independent empty cart             |
| 7    | Quantity 0, 11 or "two"         | HTTP 400; state preserved          |
| 8    | POST without CSRF token         | HTTP 403; no mutation              |
| 9    | End session, then GET           | New ID and blank customer          |
| 10   | Serialize and restore bean      | Customer/cart state preserved      |
| 11   | Search, filter and sort catalog | Matching products and price order  |
| 12   | Header Cart link → cart.html    | Same items, count and conversation |
| 13   | Increase then decrease Mouse    | Live count and exact totals update |

Verification completed: Maven clean verify succeeded; 13 JUnit tests passed; 23 real Payara HTTP/EJB checks passed. Browser checks covered the 12-product catalog, live cart count, dedicated cart route, quantity increases/decreases and totals, search, sorting, details, cross-page retention, review, remove/clear, isolated mobile session and zero JavaScript errors. Detailed output is included in `docs/VERIFICATION.md`.

## 16 Result

The application successfully meets all seven Shopping Cart requirements in the assignment. It maintains a distinct customer conversation in a real Stateful EJB, supports add/view/remove/clear operations, and displays correct contents and totals. Unit, container-integration and browser verification passed.

This is an academic cart demonstration. It does not create orders, reserve stock or process payments. Retention across server restart, timeout, clustered failover or explicit session end is not promised.

## 17 Conclusion

The project demonstrates how Stateful Session Beans preserve customer-specific conversational data while the EJB container manages lifecycle. A servlet connects the browser to the correct bean instance; immutable snapshots separate rendering from business state. Testing confirms state retention, client isolation, validation and exact cart calculations.

References: Jakarta Enterprise Beans 4.0 specification ([jakarta.ee](http://jakarta.ee)); Payara EJB documentation ([docs.payara.fish](http://docs.payara.fish)).  
Assignment: BTech\_CA3\_IT3\_AWP.pdf, Batch 2 / Group 1.

# Appendix A · Complete application source

This listing contains the authored implementation. Visual line wrapping does not alter the original source files packaged with the project.

## pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion><groupId>com.numan</groupId><artifactId>novacart</artifactId><version>2.0.0</version><packaging>war</packaging>
  <properties><maven.compiler.release>17</maven.compiler.release><project.build.sourceEncoding>UTF-8</project.build.sourceEncoding></properties>
  <dependencies><dependency><groupId>jakarta.platform</groupId><artifactId>jakarta.jakartaee-web-api</artifactId><version>10.0.0</version><scope>provided</scope></dependency><dependency><groupId>org.junit.jupiter</groupId><artifactId>junit-jupiter</artifactId><version>5.11.4</version><scope>test</scope></dependency></dependencies>
  <build><finalName>novacart</finalName><plugins><plugin><groupId>org.apache.maven.plugins</groupId><artifactId>maven-compiler-plugin</artifactId><version>3.13.0</version><configuration><compilerId>eclipse</compilerId></configuration><dependencies><dependency><groupId>org.codehaus.plexus</groupId><artifactId>plexus-compiler-eclipse</artifactId><version>2.15.0</version></dependency></dependencies></plugin><plugin><groupId>org.apache.maven.plugins</groupId><artifactId>maven-war-plugin</artifactId><version>3.4.0</version></plugin><plugin><groupId>org.apache.maven.plugins</groupId><artifactId>maven-surefire-plugin</artifactId><version>3.5.2</version></plugin></plugins></build></project>
```

## CartLine.java

```
package com.numan.novacart;
import java.io.Serializable;
import java.math.BigDecimal;
public record CartLine(Product product, int quantity) implements Serializable {
    public BigDecimal subtotal() { return product.price().multiply(BigDecimal.valueOf(quantity)); }
}
```

## CartServlet.java

```
package com.numan.novacart;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
import jakarta.servlet.ServletException;
import jakarta.json.*;
import javax.naming.InitialContext;
import javax.naming.NamingException;
import java.io.IOException;
import java.nio.charset.StandardCharsets;
import java.security.MessageDigest;
import java.util.UUID;
/** HttpSession retains the EJB proxy, while the EJB owns customer/cart state. */
@WebServlet("/api/cart")
public class CartServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;
    private ShoppingCartBean cart(HttpSession session) throws NamingException {
        ShoppingCartBean bean = (ShoppingCartBean) session.getAttribute("cartBean");
        if (bean == null) {
            InitialContext context = new InitialContext();
            try { bean = (ShoppingCartBean) context.lookup("java:module/ShoppingCartBean"); } finally {
                context.close();
            }
            session.setAttribute("cartBean", bean);
            session.setAttribute("csrf", UUID.randomUUID().toString());
        }
        return bean;
    }
    @Override protected void doGet(HttpServletRequest req, HttpServletResponse res) throws IOException, ServletException { process(req, res, false); }
    @Override protected void doPost(HttpServletRequest req, HttpServletResponse res) throws IOException, ServletException { process(req, res, true); }
    private void process(HttpServletRequest req, HttpServletResponse res, boolean mutation) throws IOException, ServletException {
        req.setCharacterEncoding("UTF-8"); res.setCharacterEncoding("UTF-8");
        res.setContentType("application/json");
    }
}
```

```

res.setHeader("Cache-Control", "no-store"); res.setHeader("X-Content-Type-Options", "nosniff");
HttpSession session = req.getSession();
// Serialize operations belonging to this browser session; other customers proceed independently.
synchronized (session) {
    try {
        ShoppingCartBean bean = cart(session);
        if (mutation) {
            String token = req.getHeader("X-CSRF-Token");
            String expected = (String) session.getAttribute("csrf");
            if (token == null || !MessageDigest.isEqual(token.getBytes(StandardCharsets.UTF_8),
expected.getBytes(StandardCharsets.UTF_8))) { error(res, 403, "Session changed. Reload the page and try
again."); return; }
            String action = req.getParameter("action");
            if (action == null) throw new IllegalArgumentException("Choose an operation.");
            switch (action) {
                case "customer" -> bean.setCustomerName(req.getParameter("name"));

                case "add" -> bean.add(req.getParameter("id"));
                case "quantity" -> bean.updateQuantity(req.getParameter("id"),
parseQuantity(req.getParameter("quantity")));
                case "remove" -> bean.remove(req.getParameter("id"));
                case "clear" -> bean.clear();
                case "end" -> { session.invalidate(); res.getWriter().write("{\"ended\":true}"); return; }
                default -> throw new IllegalArgumentException("Unknown operation.");
            }
        }
        write(res, bean.viewCart(), (String) session.getAttribute("csrf"));
    } catch (IllegalArgumentException e) { error(res, 400, e.getMessage()); }
    catch (jakarta.ejb.NoSuchEJBException e) { session.invalidate(); error(res, 409, "Your cart session
expired. Reload to begin a new session."); }
    catch (jakarta.ejb.EJBException e) {
        Throwable cause = e.getCausedByException();
        if (cause instanceof IllegalArgumentException) error(res, 400, cause.getMessage());
        else { getServletContext().log("Cart operation failed", e); error(res, 500, "Unable to update the
cart. Please try again."); }
    } catch (NamingException e) { throw new ServletException("EJB lookup failed. Deploy on a Jakarta EE
10 EJB-capable server.", e); }
}
}
private int parseQuantity(String value) { try { return Integer.parseInt(value); } catch
(NumberFormatException e) { throw new IllegalArgumentException("Enter a whole-number quantity."); } }
private void error(HttpServletResponse res, int status, String message) throws IOException {

res.setStatus(status); res.getWriter().write(Json.createObjectBuilder().add("error",
message).build().toString()); }
private void write(HttpServletResponse res, CartSnapshot s, String token) throws IOException {
    JSONArrayBuilder lines = Json.createArrayBuilder(), catalog = Json.createArrayBuilder();
    for (CartLine line : s.items()) lines.add(product(line.product()).add("quantity",
line.quantity()).add("subtotal", line.subtotal()));
    for (Product p : Catalog.PRODUCTS) catalog.add(product(p));
    res.getWriter().write(Json.createObjectBuilder().add("customerName",
s.customerName()).add("conversationId", s.conversationId()).add("revision", s.revision()).add("items",
lines).add("itemCount", s.itemCount()).add("total", s.total()).add("catalog", catalog).add("csrfToken",
token).build().toString());
}
private JsonObjectBuilder product(Product p) { return Json.createObjectBuilder().add("id",
p.id()).add("name", p.name()).add("category", p.category()).add("description",
p.description()).add("price", p.price()); }
}

```

## CartSnapshot.java

```

package com.numan.novacart;
import java.io.Serializable;
import java.math.BigDecimal;
import java.util.List;
public record CartSnapshot(String customerName, String conversationId, long revision, List<CartLine>
items) implements Serializable {
    public int itemCount() { return items.stream().mapToInt(CartLine::quantity).sum(); }
    public BigDecimal total() { return items.stream().map(CartLine::subtotal).reduce(new
BigDecimal("0.00"), BigDecimal::add); }
}

```

# CartValidationException.java

```
package com.numan.novacart;
import jakarta.ejb.ApplicationException;
/** Expected validation failure: preserve the Stateful EJB conversation. */
@ApplicationException(rollback = false)
public class CartValidationException extends IllegalArgumentException {
    private static final long serialVersionUID = 1L;
    public CartValidationException(String message) { super(message); }
}
```

# Catalog.java

```
package com.numan.novacart;
import java.math.BigDecimal;
import java.util.List;
/** Read-only catalog. The server is the single source of truth for product prices. */
public final class Catalog {
    private Catalog() {}
    public static final List<Product> PRODUCTS = List.of(
        new Product("laptop", "NovaBook Pro 14", "Laptop", "14-inch display · 16 GB RAM · 512 GB SSD", new
BigDecimal("64990.00")),
        new Product("laptop-air", "NovaBook Air 13", "Laptop", "13-inch display · 8 GB RAM · 256 GB SSD", new
BigDecimal("47990.00")),
        new Product("laptop-studio", "NovaBook Studio 16", "Laptop", "16-inch display · 32 GB RAM · 1 TB
SSD", new BigDecimal("94990.00")),
        new Product("mouse", "Arc Wireless Mouse", "Mouse", "Precision tracking · Silent clicks · Wireless",
new BigDecimal("1299.00")),
        new Product("mouse-pro", "Orbit Precision Mouse", "Mouse", "Adjustable DPI · Ergonomic grip · USB-C
charging", new BigDecimal("2499.00")),
        new Product("keyboard", "Type Mechanical Keyboard", "Keyboard", "Tactile switches · Compact layout ·
USB-C", new BigDecimal("3499.00")),
        new Product("keyboard-mini", "Type Mini 65", "Keyboard", "65% layout · Hot-swap switches · Wireless",
new BigDecimal("4499.00")),
        new Product("headphones", "Pulse Over-Ear", "Audio", "Active noise cancellation · 40-hour battery ·
Bluetooth", new BigDecimal("5999.00")),
        new Product("earbuds", "Pulse Buds", "Audio", "Pocket-size case · Touch controls · USB-C charging",
new BigDecimal("2999.00")),

        new Product("monitor", "Canvas Display 27", "Display", "27-inch QHD · IPS panel · Adjustable stand",
new BigDecimal("18990.00")),
        new Product("monitor-wide", "Canvas UltraWide 34", "Display", "34-inch curved display · WQHD ·
USB-C", new BigDecimal("32990.00")),
        new Product("hub", "Link 7-in-1 Hub", "Accessory", "HDMI · USB-A · USB-C power delivery · SD reader",
new BigDecimal("2299.00"));
    public static Product get(String id) {
        return PRODUCTS.stream().filter(product -> product.id().equals(id)).findFirst()
            .orElseThrow(() -> new CartValidationException("Choose a valid product."));
    }
}
```

# Product.java

```
package com.numan.novacart;
import java.io.Serializable;
import java.math.BigDecimal;
/** Immutable server-owned catalog entry. Prices never come from the browser. */
public record Product(String id, String name, String category, String description, BigDecimal price)
implements Serializable {}
```

# SessionCleanup.java

```
package com.numan.novacart;
import jakarta.servlet.annotation.WebListener;
import jakarta.servlet.http.*;
@WebListener
public class SessionCleanup implements HttpSessionListener {
    @Override public void sessionDestroyed(HttpSessionEvent event) {
        Object bean = event.getSession().getAttribute("cartBean");
        if (bean instanceof ShoppingCartBean cart) {
            try { cart.close(); } catch (jakarta.ejb.NoSuchEJBException ignored) { /* Already expired. */ }
        }
    }
}
```

# ShoppingCartBean.java

```
package com.numan.novacart;
import jakarta.ejb.Stateful;
import jakarta.ejb.StatefulTimeout;
import jakarta.ejb.Remove;
import java.io.Serializable;
import java.util.*;
import java.util.concurrent.TimeUnit;
/** One container-managed instance per customer conversation; never a static cart. */
@Stateful
@StatefulTimeout(value = 35, unit = TimeUnit.MINUTES)
public class ShoppingCartBean implements Serializable {
    private static final long serialVersionUID = 1L;
    private String customerName = "";
    private final String conversationId = UUID.randomUUID().toString();
    private final Map<String, Integer> quantities = new LinkedHashMap<>();
    private long revision;
    public void setCustomerName(String name) {
        if (name == null || name.strip().isEmpty() || name.strip().length() > 60 ||
name.chars().anyMatch(Character::isISOControl)) throw new CartValidationException("Enter a customer
name between 1 and 60 characters.");
        customerName = name.strip(); revision++;
    }
    public void add(String productId) {
        requireCustomer(); Catalog.get(productId);
        int quantity = quantities.getDefault(productId, 0);

        if (quantity >= 10) throw new CartValidationException("Maximum 10 units per product.");
        quantities.put(productId, quantity + 1); revision++;
    }
    public void updateQuantity(String productId, int quantity) {
        requireCustomer(); Catalog.get(productId);
        if (!quantities.containsKey(productId)) throw new CartValidationException("Product is not in your
cart.");
        if (quantity < 1 || quantity > 10) throw new CartValidationException("Quantity must be between 1 and
10.");
        quantities.put(productId, quantity); revision++;
    }
    public void remove(String productId) {
        requireCustomer(); Catalog.get(productId);
        if (quantities.remove(productId) == null) throw new CartValidationException("Product is not in your
cart.");
        revision++;
    }
    public void clear() { requireCustomer(); quantities.clear(); revision++; }
    public CartSnapshot viewCart() {
        List<CartLine> lines = quantities.entrySet().stream().map(e -> new CartLine(Catalog.get(e.getKey()),
e.getValue())).toList();
        return new CartSnapshot(customerName, conversationId, revision, List.copyOf(lines));
    }
    private void requireCustomer() { if (customerName.isEmpty()) throw new CartValidationException("Enter
your name to start shopping."); }

    /** @Remove tells the container to destroy this conversation after the invocation. */
    @Remove public void close() { quantities.clear(); }
}
```

## web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee" version="6.0"><display-name>NovaCart</display-name>
<session-config><session-timeout>30</session-timeout><cookie-config><http-only>true</http-only></cooki
e-config><tracking-mode>COOKIE</tracking-mode></session-config><welcome-file-list><welcome-file>index.h
tml</welcome-file></welcome-file-list></web-app>
```

# index.html

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<meta name="theme-color" content="#191d26">
<meta name="description" content="Thoughtfully chosen tech for your everyday workspace. Explore the
NovaCart collection.">
<title>NovaCart – Your next great setup.</title>
<link rel="stylesheet" href="assets/style.css">
<script src="assets/artwork.js" defer>
</script>
<script src="assets/app.js" defer>
</script>
</head>
<body data-page="shop">
<div class="announcement">Built for your next big idea. <span>Meet the everyday tech collection <a
href="index.html#collection">Explore →</a>
</span>
</div>
<header class="header">
<a class="brand" href="index.html" aria-label="NovaCart home">
<span class="brand-symbol">n<span>◀</span></span>
</span>nova<span>cart</span>
</a>

<nav aria-label="Main navigation">
<a href="index.html#collection">Shop the collection</a>
<a href="index.html#approach">Our approach</a>
</nav>
<div class="header-actions">
<button class="profile-button" id="profile-button" aria-label="Edit customer name">
<span class="person-icon">◯</span>
<span id="customer-label">Your session</span>
</button>
<a class="header-cart" href="cart.html" aria-label="View cart, 0 items">
<svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="1.7"
aria-hidden="true">
<path d="M6 7h12l1 14H5L6 7Z"/>
<path d="M9 8V6a3 3 0 0 1 6 0v2"/>
</svg>Cart <span id="header-count" aria-live="polite">0</span>
</a>
</div>
</header>
<main>
<section class="hero">
<div class="hero-copy">
<p class="eyebrow">
<span>
</span> EVERYDAY TOOLS. EXTRAORDINARY IDEAS.</p>
<h1>Your next<br>great <em>setup.</em>
</h1>
<p class="hero-description">For the builders. The thinkers. The late-night creators.<br>Discover tech
that moves at your pace.</p>
<a class="primary" href="#collection">Build your workspace <span>▶</span>
</a>
<div class="hero-bottom">
<span class="hero-avatars">
<b>N</b>
<b>C</b>
<b>▶</b>
</span>
<p>Less clutter. More possibility.<br>
<strong>12 thoughtfully chosen essentials.</strong>
</p>
</div>
</div>
<div class="hero-visual">
<div class="hero-orbit">
</div>
<div class="hero-orbit second">
</div>
<span class="vertical-label">DESIGNED FOR YOUR EVERYDAY</span>
<span class="hero-spec">THE NOVABOOK PRO 14<br>
<b>Power, without the noise.</b>
</span>
</div>
```

```

<div id="hero-laptop">
</div>
<div id="hero-headphones">
</div>
<div id="hero-mouse">
</div>
<a class="hero-floating" href="#collection">
<span class="live-dot">
</span>
<div>A better day starts here.<small>Find your next essential</small>
</div>
<span></span>
</a>
<span class="hero-number">01<span>/ THE COLLECTION</span>
</span>
</div>
</section>
<div class="trust-bar" id="approach">
<span>
<i></i>
<b>Built around your day</b>
<small>Work. Create. Repeat.</small>
</span>
<span>
<i></i>

<b>Transparent pricing</b>
<small>No surprise charges</small>
</span>
<span>
<i></i>

<b>Pick up where you left off</b>
<small>A cart that remembers</small>
</span>
</div>
<section class="collection" id="collection">
<div class="section-title">
<div>
<p class="eyebrow">THE TOOLS YOU REACH FOR</p>
<h2>Make room for <em>better.</em>
</h2>
</div>
<span class="section-note">A considered collection.<br>An uncompromising everyday.</span>
</div>
<div class="shopping-toolbar">
<div class="category-tabs" aria-label="Filter by category">
<button class="category-tab active" data-category="All" aria-pressed="true">All essentials
<span>12</span>
</button>
<button class="category-tab" data-category="Laptop" aria-pressed="false">Laptops</button>
<button class="category-tab" data-category="Mouse" aria-pressed="false">Mice</button>

<button class="category-tab" data-category="Keyboard" aria-pressed="false">Keyboards</button>
<button class="category-tab" data-category="Audio" aria-pressed="false">Audio</button>
<button class="category-tab" data-category="Display" aria-pressed="false">Displays</button>
<button class="category-tab" data-category="Accessory" aria-pressed="false">Accessories</button>
</div>
<div class="search-sort">
<label class="search-box">
<svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="1.7"
aria-hidden="true">
<circle cx="10" cy="10" r="6"/>
<path d="m15 15 5 5"/>
</svg>
<input id="search" placeholder="Find your next essential" aria-label="Search products">
<button id="clear-search" aria-label="Clear search" hidden>x</button>
</label>
<label class="sort-label">
<span class="sr-only">Sort products</span>
<select id="sort">
<option value="featured">Featured</option>
<option value="price-low">Price: low to high</option>
<option value="price-high">Price: high to low</option>
<option value="name">Name: A to Z</option>
</select>
</label>
</div>
</div>
<div class="results-line">
<span id="results-count" role="status">Loading the collection...</span>
<span class="subtle">Good tools. Zero guesswork.</span>

```

```

</div>
<div class="product-grid" id="products" aria-busy="true">
</div>
<div class="no-results" id="no-results" hidden>
<span>⌕</span>
<h3>No essentials found.</h3>
<p>Try a different search or explore another category.</p>
<button class="secondary" id="reset-filters">Reset filters</button>
</div>
<div class="catalog-bottom">
<span>ALL PRICES IN INR</span>
<p>Your best work starts with the right tools.</p>
<a href="cart.html">See your selection <span>⌕</span></a>
</div>
</section>
<section class="editorial">
<div>
<p class="eyebrow">THE WAY WE SEE IT</p>
<h2>Less noise.&br>More <em>possibility.</em>
</h2>

<p>A focused collection of tools that belong on your desk.&br>Choose what works for you. Leave the rest
behind.</p>
</div>
<div class="editorial-art" id="editorial-keyboard">
</div>
<span class="editorial-index">WORKSPACE / 001</span>
</section>
</main>
<footer>
<div>
<a class="brand" href="index.html">nova<span>cart</span>
</a>
<p>Good tools. Great possibilities.</p>
</div>
<div class="footer-meta">
<p>Designed & developed by Numan Qureshi</p>
<span>AWP CA3 · Batch 2 / Group 1</span>
</div>
<span class="footer-version">THE EVERYDAY TECH COLLECTION / 2026</span>
</footer>
<div class="toast" id="toast" role="status" aria-live="polite">
</div>
<dialog id="customer-dialog">
<button class="dialog-close" data-close="customer-dialog" aria-label="Close customer form">✕</button>
<span class="dialog-kicker">A LITTLE INTRODUCTION</span>

<h2>Make yourself<br>at home.</h2>
<p class="muted">Enter your name to start your personal shopping session.</p>
<form id="customer-form">
<label for="customer-name">Your name</label>
<input id="customer-name" name="name" placeholder="e.g. Numan" maxlength="60" autocomplete="name"
required>
<p class="form-error" id="name-error" role="alert">
</p>
<button class="primary" type="submit">Start shopping <span>⌕</span>
</button>
</form>
<p class="dialog-note">Your cart stays with you throughout this session.</p>
</dialog>
<dialog id="product-dialog">
<button class="dialog-close" data-close="product-dialog" aria-label="Close product details">✕</button>
<div id="product-detail">
</div>
</dialog>
<dialog id="review-dialog">
<button class="dialog-close" data-close="review-dialog" aria-label="Close cart review">✕</button>
<span class="dialog-kicker">YOUR SETUP, ALL TOGETHER</span>
<h2>Your cart,<br>reviewed.</h2>
<p id="review-customer" class="muted">
</p>
<div id="review-lines">

</div>
<div class="review-total">
<span>Cart total</span>
<b id="review-total">
</b>
</div>
<p class="dialog-note">This demo reviews your cart. No order is placed and no payment is collected.</p>
<button class="primary" data-close="review-dialog">Back to your cart <span>⌕</span>

```

```
</button>
</dialog>
</body>
</html>
```

## cart.html

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<meta name="theme-color" content="#191d26">
<meta name="description" content="Thoughtfully chosen tech for your everyday workspace. Explore the
NovaCart collection.">
<title>Your Cart – NovaCart</title>
<link rel="stylesheet" href="assets/style.css">
<script src="assets/artwork.js" defer>
</script>
<script src="assets/app.js" defer>
</script>
</head>
<body data-page="cart">
<div class="announcement">Built for your next big idea. <span>Meet the everyday tech collection <a
href="index.html#collection">Explore →</a>
</span>
</div>
<header class="header">
<a class="brand" href="index.html" aria-label="NovaCart home">
<span class="brand-symbol">N</span></a>
</span>nova<span>cart</span>
</a>

<nav aria-label="Main navigation">
<a href="index.html#collection">Shop the collection</a>
<a href="index.html#approach">Our approach</a>
</nav>
<div class="header-actions">
<button class="profile-button" id="profile-button" aria-label="Edit customer name">
<span class="person-icon">O</span>
<span id="customer-label">Your session</span>
</button>
<a class="header-cart" href="cart.html" aria-label="View cart, 0 items">
<svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="1.7"
aria-hidden="true">
<path d="M6 7h12l1 14H5L6 7Z"/>
<path d="M9 8V6a3 3 0 0 1 6 0v2"/>
</svg>Cart <span id="header-count" aria-live="polite">0</span>
</a>
</div>
</header>
<main class="cart-page">
<div class="breadcrumb">
<a href="index.html">Home</a>
<span></span>
<a href="index.html#collection">Collection</a>
<span></span>
<b>Your cart</b>
</div>
<div class="cart-page-title">
<div>
<p class="eyebrow">YOUR NEXT SETUP, COMING TOGETHER</p>
<h1>Your <em>cart.</em>
<span id="page-count">0 items</span>
</h1>
<p id="cart-greeting">Your essentials, all in one place.</p>
</div>
<a class="back-link" href="index.html#collection">← Continue exploring</a>
</div>
<div class="cart-layout">
<section class="cart-content" aria-label="Cart products">
<div class="cart-table-head">
<span>PRODUCT</span>
<span>QUANTITY</span>
<span>SUBTOTAL</span>
</div>
<div id="cart-items" aria-live="polite">
</div>
```

```

<div class="cart-tools">
<span>
<i class="live-dot">
</i> Your selection is saved for this session</span>
<button id="clear-cart" class="text-button" disabled>Clear cart</button>

</div>
</section>
<aside class="order-summary">
<span class="summary-kicker">THE BIG PICTURE</span>
<h2>Your summary</h2>
<div class="summary-row">
<span>Items <span id="summary-count">(0)</span>
</span>
<b id="subtotal">₹0.00</b>
</div>
<div class="summary-row">
<span>Additional charges</span>
<span>₹0.00</span>
</div>
<div class="summary-total">
<span>Cart total</span>
<strong id="total">₹0.00</strong>
</div>
<button class="primary" id="review" disabled>Review your selection <span>^</span>
</button>
<p class="summary-note">Cart demonstration · No payment collected</p>
<div class="summary-benefit">
<span>◊</span>
<div>Every detail, accounted for.<small>Clear prices. A cart that remembers.</small>
</div>

</div>
</aside>
</div>
<div class="cart-lower">
<div class="session-box">
<button class="session-toggle" id="session-toggle" aria-expanded="false"
aria-controls="session-detail">
<span>
<i class="live-dot">
</i> Your shopping session</span>
<span>+</span>
</button>
<div id="session-detail" hidden>
<p>Your customer name and product selections are maintained by the same Stateful Session Bean across
pages and refreshes.</p>
<dl>
<dt>Customer</dt>
<dd id="session-customer"></dd>
<dt>Conversation ID</dt>
<dd id="conversation">Connecting...</dd>
<dt>State revision</dt>
<dd id="revision">0</dd>
</dl>
<button class="text-button" id="end-session">End session & start fresh</button>
</div>

</div>
<div class="cart-tip">
<span>^</span>
<p>
<b>Make it your own.</b>
<br>Increase, decrease or remove an item.<br>Your totals update with every change.</p>
</div>
</div>
<section class="recommended">
<div class="recommended-heading">
<div>
<p class="eyebrow">A FEW GOOD ADDITIONS</p>
<h2>Round out your setup.</h2>
</div>
<a href="index.html#collection">Explore all essentials ^</a>
</div>
<div class="product-grid" id="recommendations">
</div>
</section>
</main>
<footer>
<div>
<a class="brand" href="index.html">nova<span>cart</span>
</a>

```

```

<p>Good tools. Great possibilities.</p>
</div>
<div class="footer-meta">
<p>Designed & developed by Numan Qureshi</p>
<span>AWP CA3 · Batch 2 / Group 1</span>
</div>
<span class="footer-version">THE EVERYDAY TECH COLLECTION / 2026</span>
</footer>
<div class="toast" id="toast" role="status" aria-live="polite">
</div>
<dialog id="customer-dialog">
<button class="dialog-close" data-close="customer-dialog" aria-label="Close customer form">x</button>
<span class="dialog-kicker">A LITTLE INTRODUCTION</span>
<h2>Make yourself<br>at home.</h2>
<p class="muted">Enter your name to start your personal shopping session.</p>
<form id="customer-form">
<label for="customer-name">Your name</label>
<input id="customer-name" name="name" placeholder="e.g. Numan" maxlength="60" autocomplete="name"
required>
<p class="form-error" id="name-error" role="alert">
</p>
<button class="primary" type="submit">Start shopping <span>,</span>
</button>
</form>
<p class="dialog-note">Your cart stays with you throughout this session.</p>
</dialog>

<dialog id="product-dialog">
<button class="dialog-close" data-close="product-dialog" aria-label="Close product details">x</button>
<div id="product-detail">
</div>
</dialog>
<dialog id="review-dialog">
<button class="dialog-close" data-close="review-dialog" aria-label="Close cart review">x</button>
<span class="dialog-kicker">YOUR SETUP, ALL TOGETHER</span>
<h2>Your cart,<br>reviewed.</h2>
<p id="review-customer" class="muted">
</p>
<div id="review-lines">
</div>
<div class="review-total">
<span>Cart total</span>
<b id="review-total">
</b>
</div>
<p class="dialog-note">This demo reviews your cart. No order is placed and no payment is collected.</p>
<button class="primary" data-close="review-dialog">Back to your cart <span>,</span>
</button>
</dialog>
</body>
</html>

```

# artwork.js

```
const baseArtwork = {
  laptop: `<svg viewBox="0 0 460 310" xmlns="http://www.w3.org/2000/svg"
aria-hidden="true"><defs><linearGradient id="screen" x2="1" y2="1"><stop stop-color="#192e28"/><stop
offset="1" stop-color="#628363"/></linearGradient></defs><rect x="68" y="28" width="322" height="215"
rx="12" fill="#343d37"/><rect x="78" y="39" width="302" height="191" rx="5" fill="url(#screen)"/><path
d="M80 204Q175 25 380 90V230H80Z" fill="#9bba83" opacity=".8"/><path d="M78 173Q240 110 380 44V127Q280
199 78 216Z" fill="#d2dfa9" opacity=".6"/><path d="M68 242H390L446 279Q449 287 431 290H260Q12 288 20
279Z" fill="#bec6bd"/><path d="M84 249H375L399 270H59Z" fill="#6f7b70"/><path d="M192 275H266L277
284H182Z" fill="#96a394"/><circle cx="229" cy="34" r="2" fill="#92ad85"/><path d="M90 255H375M76
261H386M120 249L104 270M159 249L149 270M200 249L195 270M240 249L240 270M282 249L287 270M325 249L336
270" stroke="#a2afa0" stroke-width="2"/></svg>`,
  mouse: `<svg viewBox="0 0 260 220" xmlns="http://www.w3.org/2000/svg" aria-hidden="true"><ellipse
cx="137" cy="185" rx="56" ry="12" fill="#000" opacity=".09"/><g transform="rotate(-18 130 110)"><path
d="M77 94Q77 34 130 32Q183 34 183 94V146Q182 190 130 194Q77 190 77 146Z" fill="#d8d2c5"
stroke="#b9b4a6" stroke-width="2"/><path d="M130 33V105M79 103Q130 119 181 103" fill="none"
stroke="#a7a799" stroke-width="2"/><rect x="123" y="59" width="14" height="32" rx="7"
fill="#586653"/><path d="M84 145Q88 180 122 184" fill="none" stroke="#ebe7dd" stroke-width="5"/><circle
cx="130" cy="149" r="3" fill="#a3ae93"/></g></svg>`,
  keyboard: `<svg viewBox="0 0 340 220" xmlns="http://www.w3.org/2000/svg" aria-hidden="true"><g
transform="rotate(-10 170 110)"><rect x="20" y="70" width="302" height="109" rx="12"
fill="#939f8d"/><rect x="20" y="63" width="302" height="107" rx="9" fill="#d4ddcb" stroke="#a9b6a1"
stroke-width="2"/><${[0,1,2,3].map((r) => Array.from({length: 11}, (_, c) => `<rect x="${30+c*26}" y="${73+r*21}"
width="21" height="16" rx="3" fill="${c===10?'#67875b':'#f1f3e9'}"
stroke="#b6c3aa"/>`)).join('')).join('')}<rect x="107" y="137" width="125" height="16" rx="3"
fill="#e8eddf"/><circle cx="311" cy="67" r="3" fill="#507248"/></g></svg>`
};

const headphones = `<svg viewBox="0 0 300 270" xmlns="http://www.w3.org/2000/svg"
aria-hidden="true"><ellipse cx="150" cy="241" rx="84" ry="13" fill="#111" opacity=".08"/><path d="M65
146V120a85 85 0 0 1 170 0v26" fill="none" stroke="#292e38" stroke-width="24"/><path d="M65 128V118a85
85 0 0 1 170 0v10" fill="none" stroke="#6e7685" stroke-width="8"/><g transform="rotate(-13 76
166)"><rect x="49" y="124" width="53" height="98" rx="23" fill="#1f2632"/><rect x="67" y="128"
width="37" height="90" rx="18" fill="#4a5362"/><path d="M76 138v70" stroke="#7e8998"
stroke-width="2"/></g><g transform="rotate(13 223 166)"><rect x="198" y="124" width="53" height="98"
rx="23" fill="#1f2632"/><rect x="196" y="128" width="37" height="90" rx="18" fill="#4a5362"/><circle
cx="222" cy="174" r="11" fill="none" stroke="#8a97a7"/><path d="m218 175 8-6" stroke="#bbc6d3"
stroke-width="2"/></g></svg>`;

const monitor = `<svg viewBox="0 0 360 260" xmlns="http://www.w3.org/2000/svg"
aria-hidden="true"><ellipse cx="180" cy="239" rx="100" ry="10" fill="#111" opacity=".08"/><path d="M163
177h34l9 49h-52Z" fill="#7c8490"/><path d="M131 222h98l15 15H116Z" fill="#b6bec6"/><rect x="24" y="28"
width="312" height="169" rx="9" fill="#333c48"/><rect x="32" y="36" width="296" height="150" rx="3"
fill="#cad6dd"/><path d="M32 170Q190 28 328 54V186H32Z" fill="#91a7bd"/><path d="M32 117Q176 126 328
37V90Q200 183 32 181Z" fill="#3e5d81"/><circle cx="180" cy="191" r="2" fill="#eee"/></svg>`;

const earbuds = `<svg viewBox="0 0 280 250" xmlns="http://www.w3.org/2000/svg"
aria-hidden="true"><ellipse cx="144" cy="224" rx="67" ry="9" fill="#111" opacity=".08"/><rect x="73"
y="132" width="137" height="83" rx="31" fill="#e5ded3" stroke="#c9beab" stroke-width="2"/><path d="M74
155h135" stroke="#c9beab" stroke-width="2"/><circle cx="142" cy="183" r="3" fill="#a4b592"/><g
transform="rotate(-22 99 90)"><ellipse cx="99" cy="72" rx="23" ry="26" fill="#f4efe6"
stroke="#c9beab"/><rect x="98" y="70" width="14" height="60" rx="7" fill="#ede8df"/><ellipse cx="91"
cy="68" rx="6" ry="11" fill="#697064"/></g><g transform="rotate(22 182 90)"><ellipse cx="182" cy="72"
rx="23" ry="26" fill="#f4efe6" stroke="#c9beab"/><rect x="169" y="70" width="14" height="60" rx="7"
fill="#ede8df"/><ellipse cx="190" cy="68" rx="6" ry="11" fill="#697064"/></g></svg>`;

const hub = `<svg viewBox="0 0 320 230" xmlns="http://www.w3.org/2000/svg" aria-hidden="true"><ellipse
cx="145" cy="185" rx="96" ry="10" fill="#111" opacity=".08"/><path d="M214 103c41-70 64-8 50 11"
fill="none" stroke="#5a6069" stroke-width="6"/><rect x="252" y="111" width="22" height="38" rx="5"
fill="#828891"/><rect x="257" y="143" width="12" height="9" rx="2" fill="#b1b8bf"/><g
transform="rotate(-12 145 127)"><rect x="42" y="93" width="195" height="71" rx="12"
fill="#8d949e"/><rect x="42" y="86" width="195" height="63" rx="12" fill="#bbc3cd"/><path d="M69
102h34v12H69zm48 0h34v12h-34zm48 0h39v12h-39Z" fill="#424c59"/><rect x="85" y="152" width="58"
height="5" rx="2" fill="#424c59"/><rect x="158" y="152" width="27" height="5" rx="2"
fill="#424c59"/><path d="m132 129 8-5" stroke="#606c7b" stroke-width="3"/></g></svg>`;

window.NovaArtwork = Object.freeze({
  laptop: baseArtwork.laptop.replaceAll('#9bba83', '#b0b9cd').replaceAll('#d2dfa9', '#e3cfac').replaceAll(
    '#192e28', '#19202e').replaceAll('#628363', '#667494'),
  'laptop-air': baseArtwork.laptop.replaceAll('#9bba83', '#c8aa8a').replaceAll('#d2dfa9', '#edd1b1'),
  'laptop-studio': baseArtwork.laptop.replaceAll('#9bba83', '#9390bd').replaceAll('#d2dfa9', '#cec8ea'),
  mouse: baseArtwork.mouse,
  'mouse-pro': baseArtwork.mouse.replaceAll('#d8d2c5', '#4b515b').replaceAll('#ebe7dd', '#727c8c').replace
    All('#b9b4a6', '#353c48'),
  keyboard: baseArtwork.keyboard.replaceAll('#d4ddcb', '#cbd1de').replaceAll('#67875b', '#d67a54'),
  'keyboard-mini': baseArtwork.keyboard.replaceAll('#d4ddcb', '#d8c9b9').replaceAll('#67875b', '#80695d'),
  headphones, earbuds, monitor,
  'monitor-wide': monitor.replaceAll('#3e5d81', '#7d668d').replaceAll('#91a7bd', '#baal7c'),
  hub
});
```

# style.css

```
:root{--ink:#191d26;
--muted:#73757a;
--paper:#f6f5f1;
--line:#e3e1da;
--accent:#ef613e;
--orange:#ff754f;
--serif:Georgia,'Times New Roman',serif}
*{box-sizing:border-box}
html{scroll-behavior:smooth;
scroll-padding-top:110px}
body{margin:0;
color:var(--ink);
background:var(--paper);
font:14px/1.5 Arial,Helvetica,sans-serif}
a{color:inherit;
text-decoration:none}
button,input,select{font:inherit;
color:inherit}
a,button,summary,select{cursor:pointer}
button:disabled{opacity:.45;
cursor:not-allowed}
a:focus-visible,button:focus-visible,input:focus-visible,select:focus-visible{outline:3px solid
var(--accent);
outline-offset:4px}
button{transition:background .18s,transform .18s}

button:hover:not(:disabled){filter:brightness(.97)}
[hidden]{display:none!important}
.announcement{height:32px;
background:var(--ink);
color:#f2f0e9;
text-align:center;
font-size:10px;
letter-spacing:.5px;
display:flex;
align-items:center;
justify-content:center;
gap:30px}
.announcement span{color:#bbc0c9}
.announcement a{color:#ff9b7d;
margin-left:10px}
.header{position:sticky;
top:0;
z-index:12;
background:var(--paper);
height:91px;
max-width:1440px;
padding:0 64px;
display:flex;
align-items:center;
justify-content:space-between;

margin:auto;
border-bottom:1px solid var(--line)}
.brand{display:inline-flex;
align-items:center;
font-size:26px;
letter-spacing:-1.3px;
font-weight:700}
.brand>span:not(.brand-symbol){font-weight:400}
.brand sup{font-size:9px;
align-self:flex-start;
margin:5px 0 0 5px}
.brand-symbol{display:inline-flex;
align-items:center;
justify-content:center;
background:var(--ink);
color:white;
width:33px;
height:33px;
border-radius:8px;
font-size:28px;
letter-spacing:-4px;
margin-right:9px}
.brand-symbol>span{font-size:15px;
color:var(--orange);
margin-top:-6px;
```

```

    margin-left:2px;
    letter-spacing:0}
nav{display:flex;
  gap:30px;
  font-size:12px}
nav a{padding:8px 0}
nav a:hover{color:var(--accent)}
.header-actions{display:flex;
  gap:26px;
  align-items:center}
.profile-button{display:flex;
  gap:8px;
  align-items:center;
  border:0;
  background:none;
  font-size:11px}
.person-icon{font-size:17px}
.header-cart{display:flex;
  gap:8px;
  align-items:center;
  border-left:1px solid var(--line);
  padding-left:23px;
  font-size:12px;
  font-weight:600}
#header-count{background:var(--ink);

  color:white;
  min-width:24px;
  height:24px;
  display:grid;
  place-items:center;
  border-radius:50%;
  font-size:10px;
  padding:0 5px}
.hero{max-width:1312px;
  margin:auto;
  display:grid;
  grid-template-columns:.97fr 1.03fr;
  min-height:505px;
  padding:43px 0 38px;
  gap:40px}
.hero-copy{padding:23px 0}
.eyebrow{font-size:9px;
  letter-spacing:1.8px;
  font-weight:700;
  margin:0 0 20px;
  display:flex;
  align-items:center;
  gap:8px}
.eyebrow>span{width:6px;
  height:6px;

  background:var(--accent);
  border-radius:50%}
h1,h2,h3,p{margin-top:0}
.hero h1{font:normal 78px/.99 var(--serif);
  letter-spacing:-4px;
  margin:0 0 23px}
h1 em,h2 em{color:var(--accent);
  font-weight:400}
.hero-description{color:var(--muted);
  font-size:13px;
  line-height:1.8;
  margin-bottom:27px}
.primary{display:inline-flex;
  align-items:center;
  justify-content:space-between;
  gap:35px;
  background:var(--ink);
  color:#fff;
  border:0;
  border-radius:5px;
  padding:14px 19px;
  font-weight:600;
  font-size:12px;
  min-height:46px}
.primary:hover{background:#313744}
.primary span{font-size:17px}
.hero-bottom{display:flex;
  gap:12px;
  align-items:center;

```

```

    margin-top:29px}
.hero-avatars{display:flex;
padding-left:0}
.hero-avatars b{width:28px;
height:28px;
background:#d5cec2;
border:2px solid var(--paper);
display:grid;
place-items:center;
border-radius:50%;
font-size:9px;
margin-left:-7px}
.hero-avatars b:first-child{margin-left:0;
background:#d5b69a}
.hero-avatars b:last-child{background:#262b35;
color:#f9bd9e}
.hero-bottom p{font-size:9px;
color:var(--muted);
margin:0;
line-height:1.7}
.hero-bottom strong{font-weight:500;

color:var(--ink)}
.hero-visual{position:relative;
background:#e9e4da;
border-radius:10px;
overflow:hidden;
min-height:414px;
isolation:isolate}
.hero-orbit{width:360px;
height:360px;
position:absolute;
top:20px;
left:80px;
border-radius:50%;
background:#ddd4c5;
z-index:-1}
.hero-orbit.second{background:none;
border:1px solid #ccc0ab;
width:455px;
height:455px;
top:-25px;
left:33px}
.vertical-label{position:absolute;
right:20px;
top:29px;
writing-mode:vertical-rl;

font-size:8px;
letter-spacing:2px;
color:#716e68}
.hero-spec{position:absolute;
top:26px;
left:26px;
font-size:8px;
letter-spacing:1.6px;
line-height:1.9}
.hero-spec b{font:normal 13px var(--serif);
letter-spacing:0}
#hero-laptop{position:absolute;
width:87%;
left:3%;
top:60px;
transform:rotate(-8deg);
filter:drop-shadow(0 20px 12px #2b2b3527)}
#hero-headphones{position:absolute;
width:26%;
right:1%;
bottom:55px;
transform:rotate(17deg);
filter:drop-shadow(0 6px 10px #26262c22)}
#hero-mouse{position:absolute;
width:20%;

left:3%;
bottom:0;
transform:rotate(15deg)}
.hero-visual svg{width:100%;
display:block}
.hero-floating{position:absolute;
bottom:29px;
left:26%;

```

```

display:flex;
gap:11px;
align-items:center;
background:#fff9;
padding:13px 16px;
border:1px solid #ffffffa6;
border-radius:6px;
backdrop-filter:blur(10px);
font-size:10px;
box-shadow:0 5px 25px #32322e0a}
.hero-floating small{display:block;
font-size:9px;
color:var(--muted);
margin-top:3px}
.hero-floating>span:last-child{font-size:20px;
margin-left:15px}
.live-dot{display:inline-block;

flex-shrink:0;
width:6px;
height:6px;
border-radius:50%;
background:#718b62}
.hero-number{position:absolute;
left:23px;
bottom:21px;
font:normal 19px var(--serif)}
.hero-number span{font:8px Arial;
letter-spacing:1px;
margin-left:8px;
display:block;
color:#716e68}
.trust-bar{max-width:1312px;
margin:auto;
border-block:1px solid var(--line);
display:grid;
grid-template-columns:repeat(3,1fr);
padding:20px 0}
.trust-bar>span{display:grid;
grid-template-columns:26px 1fr;
column-gap:8px;
align-items:center;
padding-left:30px;

border-right:1px solid var(--line)}
.trust-bar>span:first-child{padding-left:0}
.trust-bar>span:last-child{border:0}
.trust-bar i{font-size:20px;
grid-row:span 2;
font-style:normal}
.trust-bar b{font-size:10px;
font-weight:600}
.trust-bar small{font-size:9px;
color:var(--muted)}
.collection{max-width:1312px;
margin:54px auto 46px}
.section-title{display:flex;
justify-content:space-between;
align-items:end;
margin-bottom:25px}
.section-title .eyebrow{margin-bottom:11px}
h2{font:normal 37px/1.15 var(--serif);
letter-spacing:-1.2px;
margin:0}
.section-note{font-size:10px;
text-align:right;
color:var(--muted);
line-height:1.7}
.shopping-toolbar{border-block:1px solid var(--line);

padding:15px 0 12px}
.category-tabs{display:flex;
gap:6px;
flex-wrap:wrap}
.category-tab{border:0;
background:none;
padding:9px 14px;
font-size:11px;
border-radius:4px;
color:#65666a}
.category-tab.active{background:var(--ink);
color:#fff}

```

```

.category-tab span{background:#ffffff24;
border-radius:3px;
padding:2px 5px;
font-size:9px;
margin-left:5px}
.search-sort{display:flex;
justify-content:space-between;
align-items:center;
margin-top:14px;
border-top:1px solid var(--line);
padding-top:12px}
.search-box{display:flex;
align-items:center;

gap:9px;
width:330px}
.search-box input{border:0;
background:transparent;
flex:1;
font-size:11px;
min-width:0;
padding:8px 0;
outline-offset:0}
.search-box button{border:0;
background:none;
font-size:20px;
padding:0 8px}
.sort-label select{background:transparent;
border:0;
font-size:11px;
max-width:180px;
padding:8px}
.results-line{display:flex;
justify-content:space-between;
font-size:9px;
color:var(--muted);
padding:19px 0 16px}
.subtle{color:#a0a09b}
.product-grid{display:grid;

grid-template-columns:repeat(4,minmax(0,1fr));
gap:22px}
.product-card{min-width:0}
.product-visual{position:relative;
background:#eae6dc;
height:220px;
border-radius:7px;
overflow:hidden}
.product-card:nth-child(4n+2) .product-visual{background:#e4e7e9}
.product-card:nth-child(4n+3) .product-visual{background:#e9e4e3}
.product-card:nth-child(4n) .product-visual{background:#e5e9e0}
.product-art-button{width:100%;
height:100%;
padding:25px 14px;
border:0;
background:none;
display:grid;
place-items:center}
.product-art-button svg{width:100%;
height:100%;
transition:transform .3s;
filter:drop-shadow(0 7px 5px #393b4314)}
.product-art-button:hover svg{transform:translateY(-5px) scale(1.04)}
.product-tag{position:absolute;
top:12px;

left:12px;
font-size:8px;
letter-spacing:.5px;
padding:4px 7px;
background:#fffffffa8;
border-radius:3px;
pointer-events:none}
.product-tag.highlight{background:#f9d4bd;
color:#873d21}
.product-quick{position:absolute;
right:10px;
bottom:10px;
background:#fff9;
border:1px solid #fff;
padding:6px 10px;
border-radius:4px;

```

```

    font-size:9px;
    opacity:0;
    transition:opacity .2s}
.product-visual:hover .product-quick, .product-quick:focus-visible{opacity:1}
.product-copy{padding:14px 1px 9px}
.product-topline{display:flex;
    align-items:center;
    justify-content:space-between;
    gap:7px}

.product-category{font-size:8px;
    text-transform:uppercase;
    letter-spacing:1.5px;
    color:var(--muted)}
.product-state{font-size:8px;
    color:#708461;
    display:flex;
    align-items:center;
    gap:4px}
.product-state.empty{display:none}
.product-state .live-dot{width:4px;
    height:4px}
.product-copy h3{font-size:12px;
    font-weight:600;
    margin:7px 0 4px;
    line-height:1.5}
.product-description{font-size:9px;
    color:var(--muted);
    line-height:1.7;
    min-height:31px;
    margin:0 0 15px}
.product-bottom{display:flex;
    justify-content:space-between;
    align-items:center;
    gap:8px}

.price{font-weight:600;
    font-size:13px;
    letter-spacing:-.3px}
.add-button{display:flex;
    align-items:center;
    gap:7px;
    background:transparent;
    border:1px solid #cbc9c1;
    border-radius:4px;
    padding:7px 10px;
    font-size:9px}
.add-button .plus{font-size:15px;
    line-height:1}
.add-button:hover{background:var(--ink);
    color:white;
    border-color:var(--ink)}
.no-results{text-align:center;
    padding:70px 20px;
    border:1px dashed var(--line);
    border-radius:8px}
.no-results>span{font-size:35px;
    color:var(--accent)}
.no-results h3{font-size:28px var(--serif);
    margin:12px 0}
.no-results p{font-size:12px;
    color:var(--muted)}
.secondary{background:transparent;
    border:1px solid var(--line);
    border-radius:4px;
    padding:10px 18px;
    font-size:11px}
.catalog-bottom{display:flex;
    justify-content:space-between;
    align-items:center;
    margin-top:26px;
    padding-top:20px;
    border-top:1px solid var(--line);
    font-size:9px;
    color:var(--muted)}
.catalog-bottom>span{font-size:8px;
    letter-spacing:1.4px}
.catalog-bottom p{margin:0}
.catalog-bottom a{color:var(--ink)}
.catalog-bottom a span{margin-left:9px}
.editorial{max-width:1312px;

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```

margin:0 auto 48px;
border-radius:8px;
background:#e7e9df;
padding:42px;
position:relative;

overflow:hidden;
min-height:280px}
.editorial .eyebrow{font-size:8px;
color:#64695a}
.editorial h2{font-size:43px;
letter-spacing:-1.7px}
.editorial h2 em{color:#626e55}
.editorial p:not(.eyebrow){font-size:11px;
color:#737867;
margin:16px 0 0;
line-height:1.8}
.editorial-art{position:absolute;
width:47%;
right:5%;
top:0;
transform:rotate(-6deg);
opacity:.9}
.editorial-art svg{width:100%;
filter:drop-shadow(0 16px 20px #34482a20)}
.editorial-index{position:absolute;
right:29px;
bottom:24px;
font-size:8px;
letter-spacing:2px;
color:#768065}

footer{max-width:1312px;
margin:0 auto;
border-top:1px solid var(--line);
padding:29px 0;
display:flex;
justify-content:space-between;
align-items:center;
gap:30px}
footer .brand{font-size:23px}
footer p{font-size:9px;
color:var(--muted);
margin:5px 0 0}
.footer-meta{text-align:center}
.footer-meta span{font-size:8px;
color:#96968f;
letter-spacing:.5px}
.footer-version{font-size:8px;
letter-spacing:1px;
color:#96968f}
.cart-page{max-width:1312px;
margin:0 auto;
padding-top:24px;
min-height:65vh}
.breadcrumb{display:flex;
gap:10px;

font-size:10px;
color:var(--muted);
margin-bottom:43px}
.breadcrumb b{color:var(--ink);
font-weight:400}
.cart-page-title{display:flex;
justify-content:space-between;
align-items:center;
margin-bottom:34px}
.cart-page-title .eyebrow{margin-bottom:12px}
.cart-page-title h1{font:normal 54px/1.05 var(--serif);
letter-spacing:-2px;
margin:0 0 12px;
display:flex;
gap:16px;
align-items:center;
flex-wrap:wrap}
.cart-page-title h1 em{margin-left:-8px}
.cart-page-title h1>span{font:10px Arial;
letter-spacing:0;
color:var(--muted);
border:1px solid var(--line);
border-radius:25px;
padding:8px 12px}

```

```

.cart-page-title p:last-child{font-size:12px;
  color:var(--muted);
  margin:0}
.back-link{font-size:11px}
.cart-layout{display:grid;
  grid-template-columns:minmax(0,1fr) 345px;
  gap:38px;
  align-items:start}
.cart-table-head{display:grid;
  grid-template-columns:1fr 124px 100px;
  gap:20px;
  padding:15px 0;
  border-bottom:1px solid var(--line);
  font-size:8px;
  letter-spacing:1.5px;
  color:var(--muted)}
.cart-table-head>span:last-child{text-align:right}
.cart-item{display:grid;
  grid-template-columns:minmax(0,1fr) 124px 100px;
  gap:20px;
  align-items:center;
  padding:23px 0;
  border-bottom:1px solid var(--line)}
.cart-item-product{display:flex;
  gap:19px;
  align-items:center;

  min-width:0}
.cart-item-art{width:112px;
  height:101px;
  flex-shrink:0;
  border-radius:6px;
  background:#e9e5dc;
  display:grid;
  place-items:center;
  padding:6px}
.cart-item-art svg{width:100%;
  height:100%}
.cart-item-copy .product-category{font-size:8px}
.cart-item-copy h3{font-size:13px;
  margin:5px 0}
.cart-item-copy p{font-size:10px;
  color:var(--muted);
  margin:0 0 9px}
.remove-item{border:0;
  background:none;
  font-size:9px;
  color:#8a8a83;
  padding:0;
  text-decoration:underline;
  text-underline-offset:3px}
.quantity-control{display:grid;

  grid-template-columns:32px 35px 32px;
  border:1px solid #d9d7cf;
  border-radius:5px;
  width:max-content;
  overflow:hidden}
.quantity-control button{border:0;
  background:transparent;
  height:33px;
  font-size:18px}
.quantity-control button:hover{background:#e8e5dd}
.quantity-control>span{display:grid;
  place-items:center;
  font-size:12px}
.line-subtotal{text-align:right;
  font-size:13px;
  font-weight:600}
.cart-tools{display:flex;
  justify-content:space-between;
  gap:10px;
  padding:20px 0;
  color:var(--muted);
  font-size:9px}
.cart-tools>span{display:flex;
  align-items:center;
  gap:8px}
.text-button{border:0;
  background:none;

```

```

padding:0;
font-size:10px;
text-decoration:underline;
text-underline-offset:3px;
color:var(--muted)}
.order-summary{border:1px solid var(--line);
border-radius:8px;
background:white;
padding:27px;
position:sticky;
top:110px}
.summary-kicker{display:block;
font-size:8px;
letter-spacing:1.5px;
color:var(--muted);
margin-bottom:10px}
.order-summary h2{font-size:28px;
margin-bottom:30px}
.summary-row{display:flex;
justify-content:space-between;
font-size:11px;
margin:15px 0;
color:var(--muted)}

.summary-row b{color:var(--ink);
font-weight:500}
.summary-total{border-top:1px solid var(--line);
padding-top:20px;
margin:22px 0;
display:flex;
justify-content:space-between;
align-items:center;
font-size:12px}
.summary-total strong{font-size:23px;
letter-spacing:-.8px}
.order-summary .primary{width:100%;
font-size:11px;
gap:10px;
background:var(--accent)}
.order-summary .primary:hover{background:#d95031}
.summary-note{text-align:center;
font-size:8px;
color:var(--muted);
margin:11px 0 22px}
.summary-benefit{border-top:1px solid var(--line);
padding-top:20px;
display:flex;
gap:12px;
font-size:10px;

align-items:center}
.summary-benefit>span{font-size:26px}
.summary-benefit small{display:block;
font-size:8px;
color:var(--muted);
margin-top:5px}
.empty-cart{text-align:center;
padding:60px 20px;
background:#edebe3;
border-radius:8px;
margin-top:15px}
.empty-cart>span{font-size:45px;
color:#989b8e}
.empty-cart h2{font-size:32px;
margin:15px 0}
.empty-cart p{font-size:12px;
color:var(--muted);
margin-bottom:20px}
.cart-lower{display:grid;
grid-template-columns:1fr 345px;
gap:38px;
margin:15px 0 50px}
.session-box{border:1px solid var(--line);
border-radius:6px;
max-width:100%;

height:max-content}
.session-toggle{border:0;
background:transparent;
display:flex;
align-items:center;
justify-content:space-between;

```

```

padding:16px 19px;
width:100%;
font-size:10px}
.session-toggle>span:first-child{display:flex;
gap:8px;
align-items:center}
.session-toggle>span:last-child{font-size:18px;
color:var(--muted)}
#session-detail{padding:0 19px 20px;
font-size:11px;
color:var(--muted)}
#session-detail p{line-height:1.8}
dl{display:grid;
grid-template-columns:110px minmax(0,1fr);
gap:8px;
font-size:10px;
margin-bottom:19px}
dd{margin:0;
overflow-wrap:anywhere;

font-family:monospace;
color:var(--ink)}
.cart-tip{display:flex;
align-items:center;
gap:20px;
padding:10px 16px}
.cart-tip>span{font-size:40px;
color:var(--accent)}
.cart-tip p{font-size:10px;
color:var(--muted);
margin:0;
line-height:1.8}
.cart-tip b{color:var(--ink);
font-weight:500}
.recommended{margin-bottom:54px;
border-top:1px solid var(--line);
padding-top:30px}
.recommended-heading{display:flex;
justify-content:space-between;
align-items:end;
margin-bottom:25px}
.recommended-heading .eyebrow{margin-bottom:10px}
.recommended-heading h2{font-size:30px}
.recommended-heading a{font-size:10px}
.recommended .product-visual{height:175px}

dialog{border:1px solid var(--line);
background:var(--paper);
border-radius:10px;
max-width:470px;
width:calc(100% - 32px);
max-height:90vh;
padding:35px;
color:var(--ink)}
dialog:backdrop{background:#14182285;
backdrop-filter:blur(5px)}
dialog h2{font-size:42px;
margin:15px 0}
.dialog-kicker{font-size:8px;
letter-spacing:1.8px;
color:var(--accent);
font-weight:600}
.dialog-close{position:absolute;
right:16px;
top:12px;
border:0;
background:none;
padding:3px 9px;
font-size:27px;
color:var(--muted)}
.muted{color:var(--muted);

font-size:12px;
line-height:1.8}
.dialog-note{font-size:10px;
color:var(--muted);
margin:20px 0 0;
line-height:1.8}
#customer-form{margin-top:20px}
#customer-form label{display:block;
font-size:10px;
font-weight:600;

```

```

    margin-bottom:7px}
#customer-form input{border:1px solid #ccc9c0;
  background:white;
  border-radius:5px;
  width:100%;
  padding:13px;
  outline-offset:1px}
#customer-form .primary{width:100%}
.form-error{font-size:11px;
  color:#a23827;
  min-height:14px;
  margin:8px 0}
.detail-art{height:220px;
  background:#e8e3d8;
  border-radius:6px;

  display:grid;
  place-items:center;
  margin:15px 0 20px}
.detail-art svg{width:90%;
  height:90%}
.detail-title{font-size:32px!important}
.detail-price{font-size:24px;
  font-weight:600;
  margin:20px 0}
.detail-add{width:100%}
.detail-specs{font-size:12px;
  color:var(--muted);
  list-style:none;
  padding:0;
  margin:15px 0}
.detail-specs li{padding:8px 0;
  border-bottom:1px solid var(--line)}
.detail-specs li:before{content:'>';
  margin-right:10px;
  color:var(--accent)}
#review-lines{max-height:260px;
  overflow:auto}
#review-lines>p{display:flex;
  justify-content:space-between;
  gap:15px;

  font-size:11px;
  border-bottom:1px solid var(--line);
  padding:12px 0;
  margin:0}
#review-lines b{white-space:nowrap}
.review-total{display:flex;
  justify-content:space-between;
  align-items:center;
  border-top:1px solid var(--line);
  padding-top:20px;
  margin:20px 0;
  font-size:14px}
.review-total b{font-size:23px}
.toast{position:fixed;
  left:50%;
  bottom:26px;
  transform:translateX(-50%) translateY(15px);
  padding:12px 17px;
  border-radius:6px;
  background:var(--ink);
  box-shadow:0 8px 30px #10152630;
  color:#fff;
  z-index:20;
  opacity:0;
  pointer-events:none;

  transition:opacity .2s,transform .2s;
  display:flex;
  align-items:center;
  gap:22px;
  font-size:11px;
  max-width:calc(100% - 30px);
  width:max-content}
.toast.show{opacity:1;
  transform:translateX(-50%) translateY(0);
  pointer-events:auto}
.toast.error{background:#8d3527}
.toast a{color:#ffb294;
  white-space:nowrap}
.connection-error{border:1px solid #de9d8d;

```

```

background:#fff0e8;
color:#833620;
border-radius:6px;
padding:18px;
grid-column:1/-1;
font-size:12px}
.connection-error button{margin-left:15px}
.sr-only{position:absolute;
width:1px;
height:1px;
overflow:hidden;

clip:rect(0,0,0,0)}

@media(min-width:1450px){.hero,.trust-bar,.collection,.editorial,.cart-page,footer{max-width:1280px}
}
@media(max-width:1439px){.hero,.trust-bar,.collection,.editorial,.cart-page,footer{margin-left:54px;
margin-right:54px}
.header{padding:0 54px}
.hero h1{font-size:72px}
.product-visual{height:205px}
.hero{gap:28px}
.hero-spec{top:23px;
left:22px}
}
@media(max-width:1100px){.header{padding:0 32px}
.hero,.trust-bar,.collection,.editorial,.cart-page,footer{margin-left:32px;
margin-right:32px}
.hero h1{font-size:64px}
.hero-description{font-size:12px}
.hero-orbit{left:35px}
.hero-orbit.second{left:0}
.hero-visual{min-height:390px}
.product-grid{grid-template-columns:repeat(3,minmax(0,1fr));
gap:18px}
.header-actions{gap:15px}
.profile-button{font-size:10px}

nav{gap:18px}
.cart-layout,.cart-lower{grid-template-columns:minmax(0,1fr) 300px;
gap:23px}
.cart-item,.cart-table-head{grid-template-columns:minmax(0,1fr) 100px 86px;
gap:13px}
.cart-item-art{width:78px;
height:83px}
.cart-item-product{gap:12px}
.cart-item-copy h3{font-size:11px}
.line-subtotal{font-size:11px}
.order-summary{padding:22px}
.summary-total strong{font-size:21px}
.footer-version{display:none}
}
@media(max-width:820px){nav{display:none}
.hero{gap:18px;
min-height:440px;
padding-top:30px}
.hero h1{font-size:57px}
.hero-description br{display:none}
.hero-visual{min-height:355px}
.hero-spec b{font-size:11px}
.hero-orbit{left:-30px}
.hero-orbit.second{left:-65px}
#hero-laptop{top:90px;

width:100%;
left:-6%}
.hero-floating{left:17%;
bottom:25px;
padding:10px;
gap:8px;
font-size:9px}
.hero-number{display:none}
.hero-floating>span:last-child{margin-left:0}
.hero-headphones{bottom:90px}
.trust-bar b{font-size:9px}
.trust-bar>span{padding-left:17px;
grid-template-columns:18px 1fr;
gap:5px}
.trust-bar small{font-size:8px}
.product-grid{gap:16px}
.product-visual{height:170px}
.product-description{min-height:45px}

```

```

.price{font-size:12px}
.add-button{font-size:8px;
padding:6px}
.section-title h2{font-size:33px}
.category-tab{padding:8px 10px;
font-size:10px}
.cart-layout,.cart-lower{grid-template-columns:1fr}

.order-summary{position:static}
.cart-page-title h1{font-size:48px}
.cart-tip{display:none}
.cart-lower{margin-bottom:36px}
.cart-item-art{width:110px;
height:100px}
.cart-item-copy h3{font-size:13px}
.cart-table-head,.cart-item{grid-template-columns:1fr 115px 100px;
gap:15px}
.line-subtotal{font-size:13px}
.editorial-art{top:45px;
right:1%;
width:48%}
.editorial h2{font-size:37px}
.editorial p:not(.eyebrow){max-width:50%}
.recommended .product-grid{grid-template-columns:repeat(3,1fr)}
}
@media(max-width:620px){html{scroll-padding-top:20px}
.announcement{font-size:9px;
height:29px}
.announcement span{display:none}
.header{height:75px;
padding:0 20px}
.brand{font-size:23px}
.brand-symbol{width:28px;

height:29px;
font-size:24px}
.header-actions{gap:12px}
.profile-button{padding:5px 0}
#customer-label{display:none}
.person-icon{font-size:19px}
.header-cart{font-size:11px;
padding-left:13px;
gap:6px}
#header-count{min-width:20px;
height:20px;
font-size:9px}
.hero,.trust-bar,.collection,.editorial,.cart-page,footer{margin-left:20px;
margin-right:20px}
.hero{grid-template-columns:1fr;
padding:27px 0 25px;
gap:21px}
.hero-copy{padding:0}
.hero h1{font-size:68px;
letter-spacing:-3px}
.eyebrow{font-size:8px;
letter-spacing:1.3px;
margin-bottom:19px}
.hero-description{font-size:12px;
max-width:300px}

.hero-bottom{margin-top:22px}
.hero-visual{min-height:320px}
.hero-orbit{width:300px;
height:300px;
left:20px;
top:10px}
.hero-orbit.second{width:370px;
height:370px;
left:-12px;
top:-24px}
#hero-laptop{width:90%;
left:0;
top:47px}
#hero-headphones{width:28%;
bottom:40px;
right:0}
#hero-mouse{width:20%;
left:1%;
bottom:4px}
.hero-floating{bottom:22px;
left:20%;
font-size:9px}

```

```

.hero-spec{font-size:7px;
  left:18px;
  top:18px}

.vertical-label{font-size:7px;
  right:14px}

.trust-bar{padding:16px 0;
  grid-template-columns:1fr 1fr;
  gap:12px}
.trust-bar>span{padding:0;
  border:0}
.trust-bar>span:last-child{display:none}
.trust-bar small{font-size:8px}
.trust-bar b{font-size:8px}
.trust-bar i{font-size:17px}
.collection{margin-top:33px}
.section-title{margin-bottom:20px}
.section-title h2{font-size:31px}
.section-note{display:none}
.category-tabs{gap:3px}
.category-tab{font-size:9px;
  padding:8px 9px}
.category-tab span{font-size:8px}
.search-sort{gap:15px;
  align-items:center}
.search-box{width:auto;
  flex:1;
  min-width:0}
.search-box input{font-size:10px}

.sort-label select{font-size:9px;
  max-width:118px}
.subtle{display:none}
.product-grid{grid-template-columns:repeat(2,minmax(0,1fr));
  gap:21px 13px}
.product-visual{height:157px}
.product-art-button{padding:20px 6px}
.product-quick{opacity:1;
  font-size:8px;
  padding:4px 6px;
  right:7px;
  bottom:7px}
.product-tag{font-size:6px;
  padding:3px 5px;
  top:8px;
  left:8px}
.product-copy h3{font-size:11px;
  min-height:33px;
  line-height:1.5}
.product-category{font-size:7px;
  letter-spacing:.9px}
.product-description{font-size:8px;
  min-height:40px;
  line-height:1.7;
  margin-bottom:10px}

.product-bottom{align-items:flex-start;
  gap:7px;
  flex-wrap:wrap}
.price{font-size:12px;
  align-self:center}
.add-button{font-size:8px;
  min-height:28px;
  padding:5px 7px}
.product-state{font-size:7px}
.catalog-bottom>span{display:none}
.catalog-bottom p{font-size:8px}
.catalog-bottom a{font-size:9px}
.editorial{padding:27px;
  min-height:310px}
.editorial h2{font-size:37px}
.editorial p:not(.eyebrow){max-width:100%;
  font-size:10px}
.editorial-art{top:126px;
  width:63%;
  right:-2%;
  opacity:.6}
.editorial-index{font-size:7px;
  bottom:15px;
  right:18px}
.editorial p br{display:none}

```

```

footer{flex-wrap:wrap;
  gap:17px;
  padding:25px 0}
.footer-meta{text-align:right;
  font-size:8px}
.footer-meta p{max-width:140px;
  font-size:8px}
.footer-meta span{font-size:7px}
.breadcrumb{margin-bottom:25px;
  font-size:9px}
.cart-page-title{align-items:start;
  gap:12px;
  flex-direction:column}
.cart-page-title h1{font-size:49px}
.cart-page-title .eyebrow{font-size:7px}
.back-link{font-size:10px}
.cart-table-head{grid-template-columns:1fr auto;
  padding:13px 0}
.cart-table-head>span:nth-child(2){display:none}
.cart-item{grid-template-columns:minmax(0,1fr) auto;
  gap:13px;
  padding:18px 0}
.cart-item-product{grid-column:1/-1;
  gap:14px}
.cart-item-art{width:88px;
  height:82px}
.cart-item-copy h3{font-size:12px}
.quantity-control{margin-left:102px;
  grid-template-columns:28px 32px 28px}
.quantity-control button{height:30px}
.line-subtotal{font-size:13px}
.cart-tools{font-size:8px;
  align-items:center}
.cart-tools .live-dot{width:5px;
  height:5px}
.cart-tools .text-button{font-size:9px}
.order-summary{padding:25px}
.summary-total strong{font-size:25px}
.recommended-heading{align-items:start;
  gap:15px;
  flex-direction:column}
.recommended-heading h2{font-size:29px}
.recommended .product-grid{grid-template-columns:repeat(2,1fr)}
.recommended .product-visual{height:145px}
.recommended .product-card:nth-child(4){display:none}
.recommended .product-description{display:none}
.recommended .product-copy h3{min-height:33px}
.recommended .product-copy{padding-top:11px}
dialog{padding:28px}
dialog h2{font-size:37px}

.toast{font-size:10px;
  gap:15px;
  padding:11px 14px}
.toast>span{max-width:235px}
.detail-art{height:190px}
.detail-title{font-size:28px!important}
dl{grid-template-columns:91px 1fr;
  font-size:9px}
}

@media(prefers-reduced-motion:reduce){html{scroll-behavior:auto}
*{transition:none!important}
}

```

# app.js

```
'use strict';
const $ = id => document.getElementById(id);
const pageType = document.body.dataset.page;
const money = value => new Intl.NumberFormat('en-IN', {style: 'currency', currency: 'INR',
minimumFractionDigits: 2}).format(value);
let state = null;
let category = 'All';
let search = '';
let sort = 'featured';
let busy = false;
let pendingProduct = null;
let toastTimer;
let detailProduct = null;

function node(tag, className = '', text) {
  const element = document.createElement(tag);
  if (className) element.className = className;
  if (text !== undefined) element.textContent = text;
  return element;
}

function art(id, className = '') {
  const element = node('div', className);
  // Only trusted, locally authored SVG markup is inserted here, never customer input.
  element.innerHTML = window.NovaArtwork[id] || window.NovaArtwork.hub;
  return element;
}

function button(text, className, callback, label) {
  const element = node('button', className, text);
  element.type = 'button';
  if (label) element.setAttribute('aria-label', label);
  element.addEventListener('click', callback);
  return element;
}

function toast(message, error = false, cartLink = false) {
  clearTimeout(toastTimer);
  $('toast').replaceChildren(node('span', '', message));
  if (cartLink) { const link = node('a', '', 'View cart ↗'); link.href = 'cart.html';
  $('toast').append(link); }
  $('toast').className = 'toast show' + (error ? ' error' : '');
  toastTimer = setTimeout(() => $('toast').classList.remove('show'), 4300);
}

function dialogOpen(id) { if (!$.id).open) $(id).showModal(); }
function requestName(productId = null) {
  pendingProduct = productId;
  $('customer-name').value = state?.customerName || '';
  $('name-error').textContent = '';
  dialogOpen('customer-dialog');
  $('customer-name').focus();
}

async function load() {
  try {
    const response = await fetch('api/cart', {cache: 'no-store'});
    const data = await response.json();
    if (!response.ok) throw new Error(data.error || 'Could not retrieve your cart.');
```

---

```

    body: new URLSearchParams({action, ...values})
  });
  const data = await response.json();
  if (!response.ok) throw new Error(data.error || 'Could not update your cart.');
```

```

  if (data.ended) {
    await load();
    toast('A new shopping session is ready.');
```

```

  } else {
    state = data;
    toast(message, false, action === 'add' && pageType === 'shop');
```

```

  }
  return true;
} catch (error) {
  if (action === 'customer') $('name-error').textContent = error.message;
  toast(error.message, true);
  return false;
} finally {
  busy = false;
  render();
}
}

async function addProduct(id) {
  if (busy || !state) return;
  if (!state.customerName) { requestName(id); return; }
  const product = state.catalog.find(item => item.id === id);

  if (product) await mutate('add', {id}, product.name + ' added to your cart.');
```

```

}

function tagFor(product) {
  if (product.id === 'laptop') return ['THE WORKSPACE HERO', true];
  if (product.id === 'headphones') return ['TUNE INTO YOUR DAY', true];
  if (product.id === 'keyboard-mini') return ['SMALL DESK. BIG IDEAS.', false];
  if (product.id === 'monitor-wide') return ['ROOM TO CREATE', false];
  return ['EVERYDAY ESSENTIAL', false];
}

function productCard(product) {
  const card = node('article', 'product-card');
  const visual = node('div', 'product-visual');
  const image = button('', 'product-art-button', () => showProduct(product.id), 'View details for ' +
product.name);
  image.append(art(product.id));
  const [label, highlight] = tagFor(product);
  visual.append(image, node('span', 'product-tag' + (highlight ? ' highlight' : ''), label));
  visual.append(button('Quick view >', 'product-quick', () => showProduct(product.id), 'Quick view ' +
product.name));
  const copy = node('div', 'product-copy');
  const top = node('div', 'product-topline');
  top.append(node('span', 'product-category', product.category));
  const quantity = state.items.find(item => item.id === product.id)?.quantity || 0;
  if (quantity) { const indicator = node('span', 'product-state'); indicator.append(node('i',
'live-dot'), node('span', '', quantity + ' in cart')); top.append(indicator); }

  copy.append(top, node('h3', '', product.name), node('p', 'product-description',
product.description));
  const bottom = node('div', 'product-bottom');
  const add = button('', 'add-button', () => addProduct(product.id), 'Add ' + product.name + ' to
cart');
  add.append(node('span', 'plus', '+'), node('span', '', quantity >= 10 ? 'Limit reached' : 'Add to
cart'));
  add.disabled = busy || quantity >= 10;
  bottom.append(node('span', 'price', money(product.price)), add);
  copy.append(bottom);
  card.append(visual, copy);
  return card;
}

function renderCatalog() {
  let products = state.catalog.filter(product => (category === 'All' || product.category === category)
&& (product.name + ' ' + product.category + ' ' + product.description).toLowerCase().includes(search));
  if (sort === 'price-low') products.sort((a, b) => a.price - b.price);
  if (sort === 'price-high') products.sort((a, b) => b.price - a.price);
  if (sort === 'name') products.sort((a, b) => a.name.localeCompare(b.name));
  $('products').replaceChildren(...products.map(productCard));
  $('products').setAttribute('aria-busy', 'false');
  $('results-count').textContent = 'Showing ' + products.length + ' of ' + state.catalog.length + '
essentials';
  $('no-results').hidden = products.length !== 0;
  $('clear-search').hidden = !search;
}

function renderCart() {
  const units = state.itemCount;
  $('page-count').textContent = units + (units === 1 ? ' item' : ' items');
```

```

    $('cart-greeting').textContent = state.customerName ? 'A considered selection, put together by ' +
state.customerName + ' : ' : 'Your essentials, all in one place.';
    $('summary-count').textContent = '(' + units + ')';
    $('subtotal').textContent = money(state.total);
    $('total').textContent = money(state.total);
    $('clear-cart').disabled = busy || !units;
    $('review').disabled = busy || !units;
    $('end-session').disabled = busy;
    $('session-customer').textContent = state.customerName || 'Not entered yet';
    $('conversation').textContent = state.conversationId;
    $('revision').textContent = state.revision;
    const container = $('cart-items');
    container.replaceChildren();
    if (!state.items.length) {
        const empty = node('div', 'empty-cart');
        const link = node('a', 'primary', 'Find your essentials >'); link.href = 'index.html#collection';
        empty.append(node('span', '', '◊'), node('h2', '', 'A little room for possibility.'), node('p', '',
'Your cart is empty. Let's find something that belongs on your desk.'), link);
        container.append(empty);
    }
    for (const item of state.items) {
        const row = node('article', 'cart-item');
        row.dataset.product = item.id;
        const product = node('div', 'cart-item-product');
        const copy = node('div', 'cart-item-copy');
        const remove = button('Remove item', 'remove-item', () => mutate('remove', {id: item.id}, item.name
+ ' removed.'), 'Remove ' + item.name);
        remove.disabled = busy;
        copy.append(node('span', 'product-category', item.category), node('h3', '', item.name), node('p',
'', money(item.price) + ' each'), remove);
        product.append(art(item.id, 'cart-item-art'), copy);
        const quantity = node('div', 'quantity-control');
        quantity.setAttribute('aria-label', item.name + ' quantity');
        const decrease = button('-', '', () => mutate('quantity', {id: item.id, quantity: item.quantity -
1}), 'Decrease ' + item.name + ' quantity');
        const increase = button('+', '', () => mutate('quantity', {id: item.id, quantity: item.quantity +
1}), 'Increase ' + item.name + ' quantity');
        decrease.disabled = busy || item.quantity <= 1;
        increase.disabled = busy || item.quantity >= 10;
        const count = node('span', '', item.quantity); count.setAttribute('aria-label', 'Quantity ' +
item.quantity);
        quantity.append(decrease, count, increase);
        row.append(product, quantity, node('strong', 'line-subtotal', money(item.subtotal)));
        container.append(row);
    }
    // Recommend actual catalog products not already selected, with deterministic ordering.

    const selected = new Set(state.items.map(item => item.id));
    const suggested = [...state.catalog.filter(product => ['mouse', 'keyboard', 'headphones',
'hub'].includes(product.id)), ...state.catalog];
    const ids = new Set();
    const recommendations = suggested.filter(product => !selected.has(product.id) && !ids.has(product.id)
&& ids.add(product.id)).slice(0, 4);
    $('recommendations').replaceChildren(...recommendations.map(productCard));
    document.querySelector('.recommended').hidden = recommendations.length === 0;
}
function render() {
    if (!state) return;
    $('header-count').textContent = state.itemCount;
    document.querySelector('.header-cart').setAttribute('aria-label', 'View cart, ' + state.itemCount + '
items');
    $('customer-label').textContent = state.customerName || 'Your session';
    $('profile-button').title = state.customerName ? 'Shopping as ' + state.customerName + ' - edit name'
: 'Enter your customer name';
    $('customer-form').querySelector('button').disabled = busy;
    if (pageType === 'shop') renderCatalog(); else renderCart();
    if (detailProduct && $('product-dialog').open) refreshDetailButton();
}
function refreshDetailButton() {
    const add = $('detail-add');
    if (!add) return;
    const quantity = state.items.find(item => item.id === detailProduct)?.quantity || 0;

    add.disabled = busy || quantity >= 10;
    add.querySelector('span:first-child').textContent = quantity >= 10 ? 'Maximum quantity reached' :
'Add to your cart';
}
function showProduct(id) {
    if (!state) return;
    const product = state.catalog.find(item => item.id === id);
    if (!product) return;

```

```

    detailProduct = id;
    const content = $('product-detail');
    content.replaceChildren();
    content.append(node('span', 'dialog-kicker', product.category.toUpperCase(), art(id, 'detail-art'),
node('h2', 'detail-title', product.name)));
    const specs = node('ul', 'detail-specs');
    product.description.split(' ').forEach(text => specs.append(node('li', '', text)));
    content.append(specs, node('div', 'detail-price', money(product.price)));
    const add = button('', 'primary detail-add', () => { $('product-dialog').close(); addProduct(id); });
    add.id = 'detail-add'; add.append(node('span', '', 'Add to your cart'), node('span', '', 'x'));
    content.append(add, node('p', 'dialog-note', 'Sample INR pricing. Selected products stay in your
customer session.'));
    dialogOpen('product-dialog'); refreshDetailButton();
}
function reviewCart() {
  if (!state?.itemCount) return;
  $('review-customer').textContent = 'Prepared for ' + state.customerName + ' · ' + state.itemCount + '
items';
  $('review-lines').replaceChildren();
  for (const item of state.items) { const row = node('p'); row.append(node('span', '', item.name + ' x
' + item.quantity), node('b', '', money(item.subtotal))); $('review-lines').append(row); }
  $('review-total').textContent = money(state.total);
  dialogOpen('review-dialog');
}
$('profile-button').addEventListener('click', () => { if (state) requestName(); });
$('customer-form').addEventListener('submit', async event => {
  event.preventDefault();
  const productId = pendingProduct;
  if (await mutate('customer', {name: $('customer-name').value}, 'Your personal shopping session is
ready.')) {
    $('customer-dialog').close(); pendingProduct = null;
    if (productId) await addProduct(productId);
  }
});
$('customer-dialog').addEventListener('close', () => { pendingProduct = null; });
document.querySelectorAll('[data-close]').forEach(element => element.addEventListener('click', () =>
$(element.dataset.close).close()));
if (pageType === 'shop') {
  $('hero-laptop').append(art('laptop')); $('hero-headphones').append(art('headphones'));
  $('hero-mouse').append(art('mouse')); $('editorial-keyboard').append(art('keyboard-mini'));
  document.querySelectorAll('.category-tab').forEach(tab => tab.addEventListener('click', () => {
    category = tab.dataset.category;

    document.querySelectorAll('.category-tab').forEach(item => { const active = item === tab;
item.classList.toggle('active', active); item.setAttribute('aria-pressed', active); });
    if (state) renderCatalog();
  }));
  $('search').addEventListener('input', () => { search = $('search').value.trim().toLowerCase(); if
(state) renderCatalog(); });
  $('clear-search').addEventListener('click', () => { $('search').value = ''; search = '';
renderCatalog(); $('search').focus(); });
  $('sort').addEventListener('change', () => { sort = $('sort').value; if (state) renderCatalog(); });
  $('reset-filters').addEventListener('click', () => {
    search = ''; $('search').value = ''; document.querySelector('[data-category="All"]').click();
  });
} else {
  $('clear-cart').addEventListener('click', () => { if (!busy && state?.itemCount && confirm('Remove
every product from your cart?')) mutate('clear', {}, 'Your cart is cleared. Your customer session stays
active.'); });
  $('end-session').addEventListener('click', () => { if (!busy && confirm('End this customer session
and clear its cart?')) mutate('end'); });
  $('session-toggle').addEventListener('click', () => { const expanded =
$('session-toggle').getAttribute('aria-expanded') !== 'true';
$('session-toggle').setAttribute('aria-expanded', expanded); $('session-detail').hidden = !expanded;
$('session-toggle').lastElementChild.textContent = expanded ? '-' : '+'; });
  $('review').addEventListener('click', reviewCart);
}
// A back/forward navigation or return to this tab reloads the server snapshot.

window.addEventListener('pageshow', event => { if (event.persisted) load(); });
document.addEventListener('visibilitychange', () => { if (!document.hidden && !busy && state) load();
});
load();

```

# run.ps1

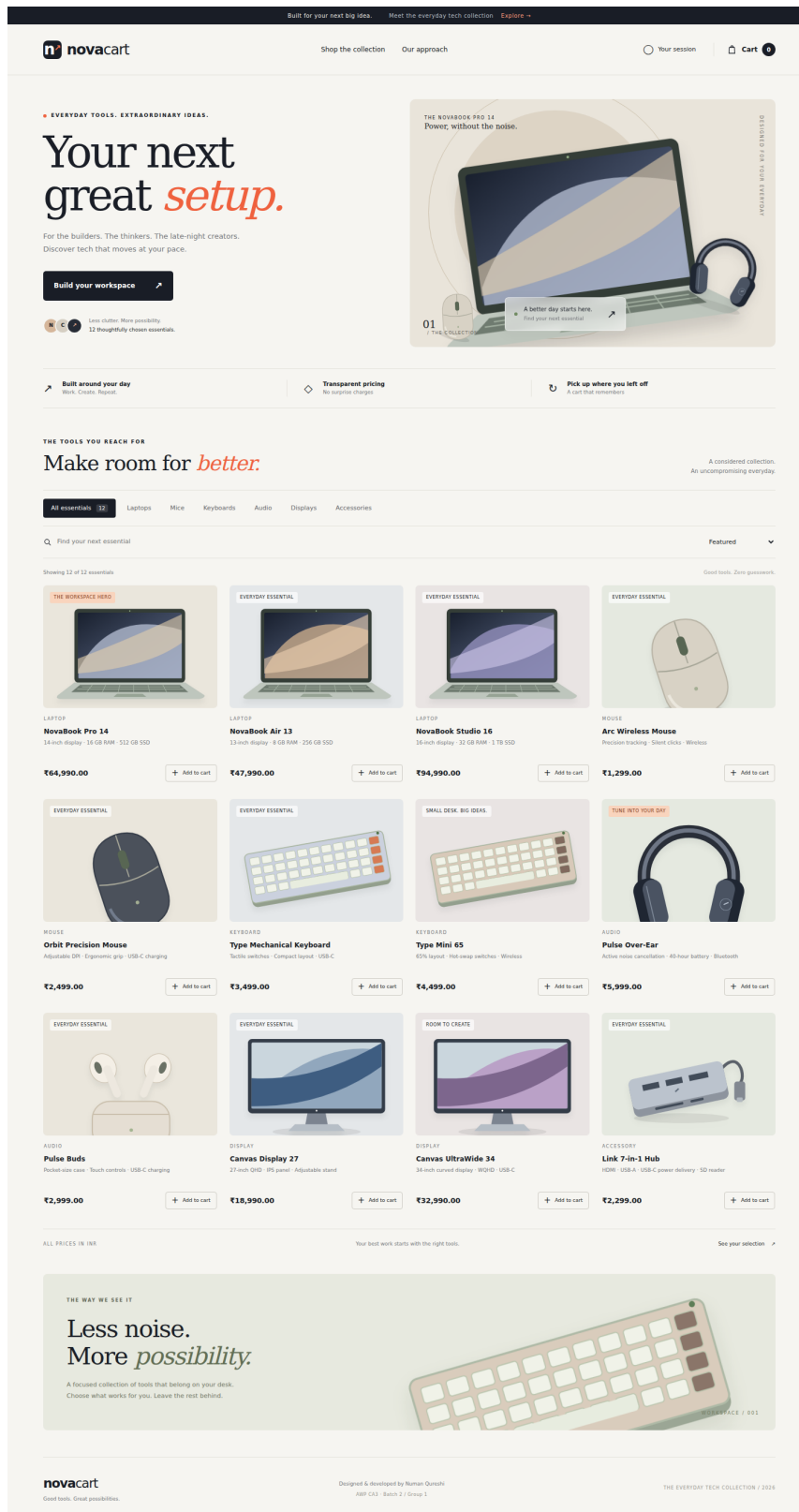
```
$ErrorActionPreference = "Stop"
Set-Location $PSScriptRoot
# Prefer the JDK 17 installed during setup; do not accidentally launch with a newer JDK.
$jdk17 = Get-ChildItem "C:\Program Files\Microsoft" -Directory -Filter "jdk-17*" -ErrorAction
SilentlyContinue | Sort-Object Name -Descending | Select-Object -First 1
if ($jdk17 -and (Test-Path (Join-Path $jdk17.FullName "bin/java.exe"))) {
    $env:JAVA_HOME = $jdk17.FullName
    $env:Path = "$env:JAVA_HOME\bin;$env:Path"
}
if (-not (Get-Command java -ErrorAction SilentlyContinue)) { throw "Install JDK 17 and add Java to
PATH, then run again." }
if (-not (Test-Path "target/novacart.war")) { if (-not (Get-Command mvn -ErrorAction SilentlyContinue))
{ throw "Install Maven 3.9+ to build source, then run mvn clean verify." }; & mvn clean verify; if
($LASTEXITCODE -ne 0) { throw "Maven build failed." } }
$javaVersion = cmd /c "java -version 2>&1" | Out-String
if ($javaVersion -notmatch 'version "17\.' ) { throw "This project requires Java 17. Install it with:
winget install --id Microsoft.OpenJDK.17 --exact; then run again." }
Write-Host "Starting with Java 17..."
New-Item -ItemType Directory -Force ".runtime" | Out-Null
$runtime = ".runtime/payara-micro.jar"
$expected = "8e3ed1276234278034a7ac94efb0400eb0d1db733e20b0dc1f5b9178de2f82ae"
if (-not (Test-Path $runtime)) { Write-Host "Downloading Payara runtime (first run only)...";
Invoke-WebRequest -Uri "https://repo.maven.apache.org/maven2/fish/payara/extras/payara-micro/6.2025.1/p
ayara-micro-6.2025.1.jar" -OutFile "$runtime.part"; Move-Item "$runtime.part" $runtime }
if ((Get-FileHash $runtime -Algorithm SHA256).Hash.ToLower() -ne $expected) { throw "Runtime checksum
mismatch. Delete .runtime/payara-micro.jar and try again." }
Write-Host "After startup, open http://localhost:8080/novacart/"
& java -jar $runtime --noCluster --port 8080 --deploy target/novacart.war --contextroot novacart
```

# run.sh

```
#!/usr/bin/env bash
set -euo pipefail
cd "$(dirname "$0")"
command -v java >/dev/null || { echo "Install JDK 17 first."; exit 1; }
if [ ! -f target/novacart.war ]; then mvn clean verify; fi
mkdir -p .runtime
if [ ! -f .runtime/payara-micro.jar ]; then
    curl -fL
https://repo.maven.apache.org/maven2/fish/payara/extras/payara-micro/6.2025.1/payara-micro-6.2025.1.jar
-o .runtime/payara-micro.jar.part
mv .runtime/payara-micro.jar.part .runtime/payara-micro.jar
fi
python3 - <<'CHECK'
import hashlib
from pathlib import Path
assert hashlib.sha256(Path('.runtime/payara-micro.jar').read_bytes()).hexdigest() ==
"8e3ed1276234278034a7ac94efb0400eb0d1db733e20b0dc1f5b9178de2f82ae", 'Runtime checksum mismatch; delete
.runtime/payara-micro.jar and retry.'
CHECK
echo "After startup, open http://localhost:8080/novacart/"
exec java -jar .runtime/payara-micro.jar --noCluster --port 8080 --deploy target/novacart.war
--contextroot novacart
```

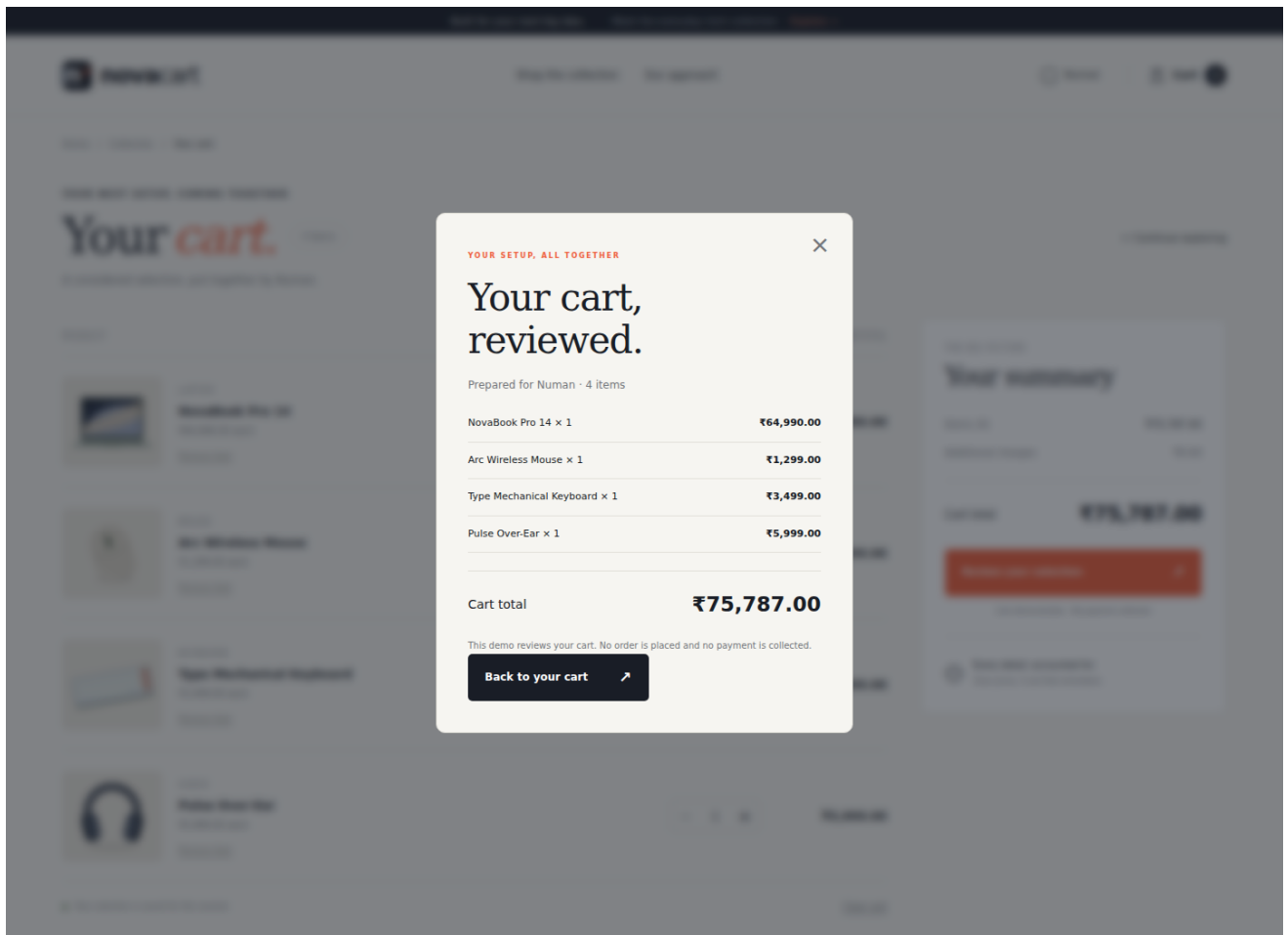
# Appendix B · Actual application output

Actual output of Version 2. The cart screenshot shows Laptop, Mouse, Keyboard and Pulse Over-Ear headphones: four units, total ₹75,787.00. Increasing Mouse to two units produces five units and ₹77,086.00. Browser navigation and refresh retain this customer conversation.



Expanded storefront: 12 products and six categories





Cart review dialog - no payment is collected